Panasonic

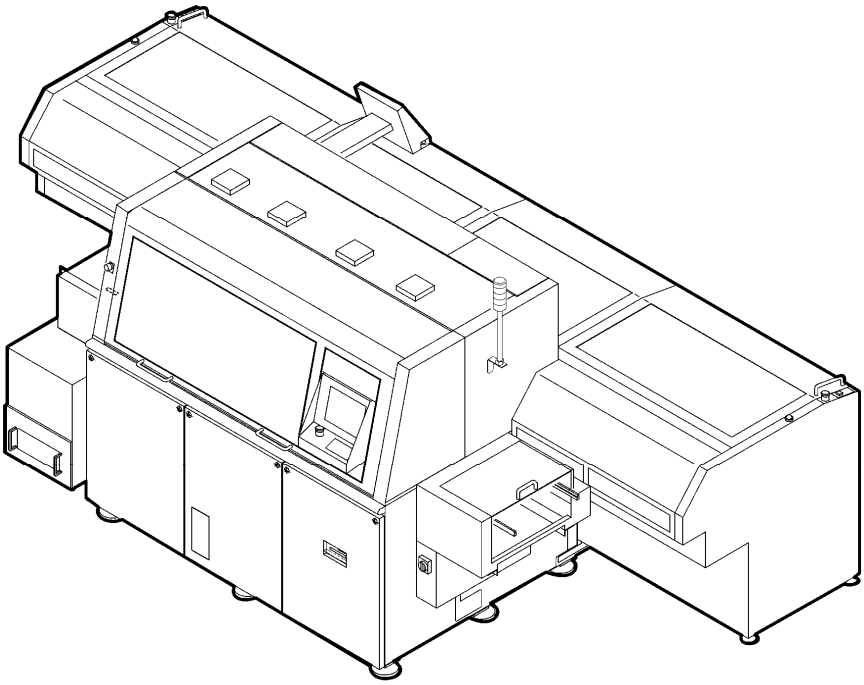
Operating Instructions

Reference Manual

High-Speed Axial Lead Component Insertion Machine

AV131

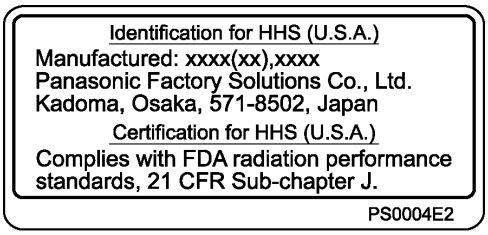
Model No. **NM-EJA2A**
NM-EJ3A



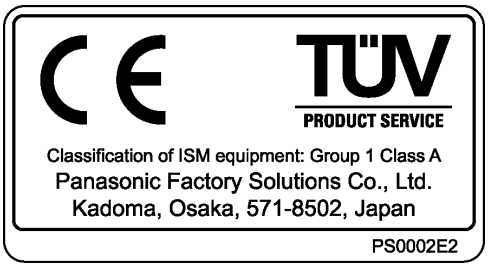
Thank you for purchasing AV131, the high-speed axial lead component insertion machine.
Before placing the machine in service, be sure to read the instruction manual for proper usage.
After that, store it carefully and read it if necessary.

N7201A234E00

Marking Samples for Product Safety



HHS-Mark for U.S.A.



CE-Mark

USAGE OF WARNING SHEETS IN MAINTENANCE

Before performing maintenance, affix the included warning sheets (see below) to prominent positions, such as the operating panel or the vicinity of it, for safety.
During the maintenance by two or more persons in particular, as well as using these warning sheets to ensure safety, be sure to send signals to each other.

This warning sheet uses a magnet. Do not use it or store it near magnetic media or any other items susceptible to magnetic influence.

Use of this machine:
This machine is designed and constructed to be used for placing electric and electronic components on PCB.
This machine shall be used for this purpose only.

It shall not be used for any other purpose.

Operation of this machine:
This machine shall be operated by professional, specially trained operators.

It shall not be operated by non-professional, not specially trained operators.

Regarding the training course, please contact Panasonic Factory Solutions Co., Ltd. or PFSE.

⚠ WARNING

Do not allow any person who has not received special training to operate the machine.

Prohibition

The machine many be operated improperly, causing injury.

- Either carry out training by yourself of consult us.

Machine was manufactured by: Panasonic Factory Solutions Co., Ltd.

441-13, Nagahasu, Tateishi-cho, Tosu, Saga 841-8585, Japan
Phone number: 81-942-84-2644
Fax number: 81-942-84-2636

For Service and Supply of Spares in Europe
you may contact Panasonic Europe:

Continental Europe
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Panasonic Factory Solutions Europe
Winsbergring 15
22525 HamburgGermany
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Bracknell, Berks. RG12 8 FP
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Year: 2008 CE

D79REC-02-010-A0

Table of Contents

1. INTRODUCTION 1-1
1.1. How to Use This Manual 1.1-1
1.1.1 Document Summary 1.1-1
1.1.2 Conventions Used 1.1-2
2. OVERVIEW 2-1
2.1. Features 2.1-1
2.1.1 Features 2.1-1
2.2. Mechanical Specifications 2.2-1
2.2.1 Machine Specifications 2.2-1
2.2.2 Insertion Component Specifications 2.2-3
2.2.3 Applicable Insertion Components 2.2-4
2.2.4 Packaging of Taped components 2.2-5
2.3. Control Specifications 2.3-1
2.3.1 Control Specifications 2.3-1
2.4. PCB Design Standard 2.4-1
2.4.1 PCB Specifications 2.4-1
2.4.2 Restrictions on Height of Components Already Mounted on Top and Bottom Sides of PCB 2.4-2
2.4.3 PCB Accuracies 2.4-3
2.5. Insertion Restrictions 2.5-1
2.5.1 Insertion Restrictions 2.5-1
3. VARIOUS FUNCTIONS 3-1
3.1. Multi-Interlock Function 3.1-1
3.1.1 Multi-Interlock Function 3.1-1
3.2. Recovery Data Compensation Function 3.2-1
3.2.1 Recovery Data Compensation Function 3.2-1
3.3. Backlight Setting 3.3-1
3.3.1 Backlight Setting 3.3-1
3.4. Help Information 3.4-1
3.4.1 Help Information 3.4-1
4. SCREEN TRANSITION 4-1
4.1. Screen Transition Diagrams 4.1-1
4.1.1 Home 4.1-1
4.1.2 Production 4.1-2

4.1.3 Production Configuration 4.1-3
4.1.4 Performance Information 4.1-4
4.1.5 Data Modify 4.1-8
4.1.6 Production Data Teaching 4.1-15
4.1.7 File Managing 4.1-16
4.1.8 Machine Adjustment 4.1-17
4.1.9 Machine Parameter 4.1-18
4.1.10 Machine Configuration 4.1-19
4.1.11 Maintenance 4.1-20

5. PRODUCTION MANAGEMENT INFORMATION 5-1

5.1. Performance Information 5.1-1
5.1.1 Data Items on the Performance Information Screen 5.1-2
5.1.2 Performance Information (Machine) 5.1-4
5.1.3 Performance Information (Model) 5.1-5
5.2. NC Data Information 5.2-1
5.2.1 Data Information 5.2-1
5.3. Feeder Information 5.3-1
5.3.1 Feeder Information 5.3-1
5.4. Rotary Chuck Information 5.4-1
5.4.1 Data Items on the Rotary Chuck Information Screen 5.4-1
5.4.2 Displaying the Rotary Chuck Information 5.4-2

6. SYSTEM MANAGEMENT 6-1

6.1. Initial Setting (Overview) 6.1-1
6.1.1 Overview 6.1-1
6.2. Production Configuration 6.2-1
6.3. Machine Parameter 6.3-1
6.4. Machine Configuration 6.4-1

1. INTRODUCTION

This chapter describes how to use this manual.

1.1. How to Use This Manual

D79REC-06-010-A0

1.1.1 Document Summary

This section describes how the AV131 Operating Instructions are organized.

Operation Manual
Describes the guidelines for basic operations.

1. INTRODUCTION
2. SAFETY PRECAUTIONS
3. DESCRIPTION
4. START/STOP
5. PROCEDURE FOR PRODUCTION (BASICS)
6. TROUBLESHOOTING

Programming Manual
Discusses the method of creating various programs.

1. INTRODUCTION
2. DATA CREATION
3. NC DATA CREATION
4. PART DATA CREATION
5. P.C.B. DATA CREATION
6. P.C.B. RECOGNITION DATA CREATION
7. DATA CHECK PROCEDURE
8. FILE MANAGING

Reference Manual
Offers helpful information clarifying unclear points you may encounter in operating the equipment.

1. INTRODUCTION
2. OVERVIEW
3. VARIOUS FUNCTIONS
4. SCREEN TRANSITION
5. PRODUCTION MANAGEMENT INFORMATION
6. SYSTEM MANAGEMENT

Maintenance Manual
Provides essential information for performing routine maintenance.

1. INTRODUCTION
2. SAFETY PRECAUTIONS
3. INSTALLATION
4. CONTROL SYSTEM
5. SENSOR LAYOUT
6. PREPARED MATERIALS
7. INSPECTION
8. MAINTENANCE GUIDE
9. SYSTEM CHECK FUNCTIONS
10. RECYCLING
11. APPENDIX

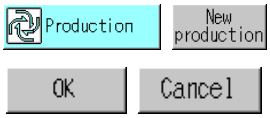
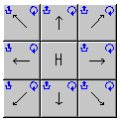
Error Code List
Contains organized error codes to be displayed when troubles occur.

1. INTRODUCTION
2. ERROR CODES

1.1.2 Conventions Used

■ Buttons, Keys and Switches

For buttons, keys and switches, this manual uses the following conventions for description of the procedure except for the functions or roles of this machine.

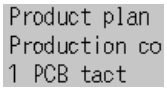
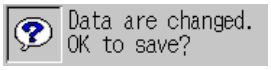
Name	Explanation	Indication example	Symbol	Example of use
Button	Pressed on the screen.		[ ]	[Production], [New production], [OK], [Cancel], etc.
Key	Used to input characters and numbers. (Keyboard)			[ESC], [BS], [ENT], etc.
	Used to move the camera etc. (Teaching keys)			[H], etc.
Switch	Used in operation other than on the screen (e.g., on control panel, sub-control panel, power-control panel).		< >	<START>, <STOP>, <ENABLING>, <PARTS EXCHANGE>, etc.

(Note) < ENABLING > + <START> indicates that you should press <START> with < ENABLING > held down.

Icons or buttons are used to show keys with no name, etc. (e.g. function keys, scroll buttons).

Function key	
Scroll button	

■ Screen Display

Item	Indication example	Symbol	Example of use
Screen name or on-screen item		“ ”	“Product plan”
Message			“Data are changed. OK to save?”

■ Cautionary Statements

Cautionary statements and cross-references are entitled or marked with an icon as follows:

Notation	Meaning
=REMARKS=	Matters requiring attention in operation
=HINT=	Helpful suggestions or available operations other than those described here
=📖=	Supplementary information for learning more about operations

These titles may be numbered as ‘=HINT=1’ in figures and footnoted.

■ Abbreviations

The names of the referential manuals are abbreviated as follows:

Ope.: Operation Manual Ref.: Reference Manual
Prg.: Programming Manual Mtn.: Maintenance Manual

The manuals are cross-referenced with the notation ‘Manual Name / Chapter Title / Section Title’.
Ex.) =📖= Refer to ‘Ope. / VARIOUS FUNCTIONS / Move Axis’ for details.

2. OVERVIEW

This chapter describes the outline of the AV131.

2.1. Features

D79REC-12-010-A0

The AV131 is a VCD axial lead component insertion machine capable of high-speed insertion at a tact time of 0.12 s/component, providing high economic efficiency.
With the adoption of vertical feeders that allows multiple inputs (120 inputs) and the two separate supply units that improve operation rate, the AV131 is promising for greatly contributing to continuous unmanned operation.
Also, the optionally available jumper wire insertion function enables insertion of jumper wires without the need for taping, which not only eliminates waste but also realizes highly efficient and flexible production.

2.1.1 Features

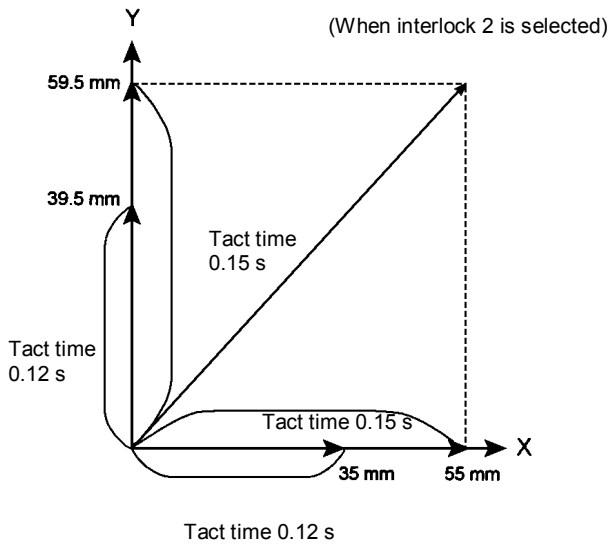
■ Higher Productivity

◆ Insertion Tact Time

- High-speed insertion at 0.12 s/component improves the productivity significantly.

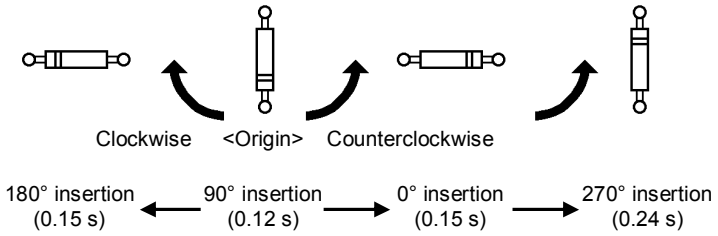
◆ High-speed X-Y Table

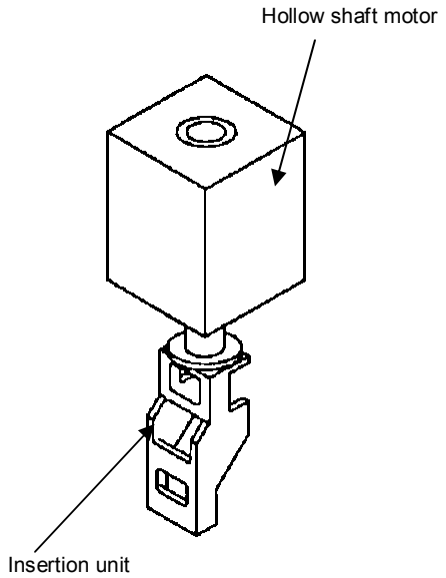
- Lightweight and highly rigid design provides higher speed X-Y table.



◆ Insertion Tact Time for Four Directions (0, 90, 180 and 270 degrees)

- Compact and lightweight insertion unit and the direct drive system adopting a hollow shaft motor offer high-speed insertion while the insertion unit is rotating.





◆ Two Separate Part Supply Units (NM-EJA3A)

- Part supply unit separated into two tables allows the supply operation mode to be selected according to the production needs, thus improving operation rate. (5 modes are available.)

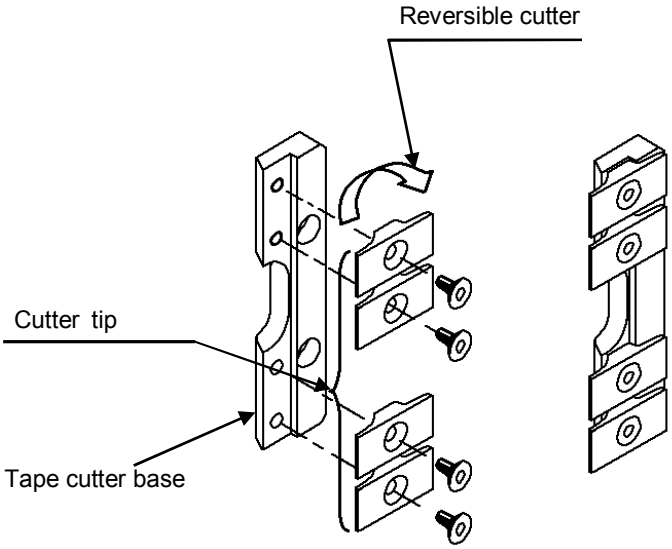
◆ Random-Access System

- The random-access system allows insertion errors to be recovered quickly by just operating the buttons.

■ Higher Reliability

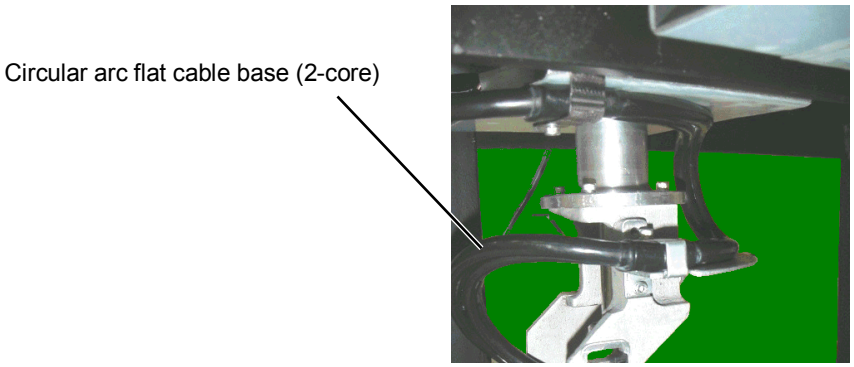
◆ Longer Life Consumable Parts

- Reversible tape cutter and reversible lead cutter (fixed side) offer a longer life of these consumable parts.
- The insertion guide and pusher made from carbide materials also ensure a longer life of them.



◆ Circular Arc Flat Cable

- The head and the anvil rotation unit adopting the circular arc flat cable offer a longer life for the wiring in the movable section.

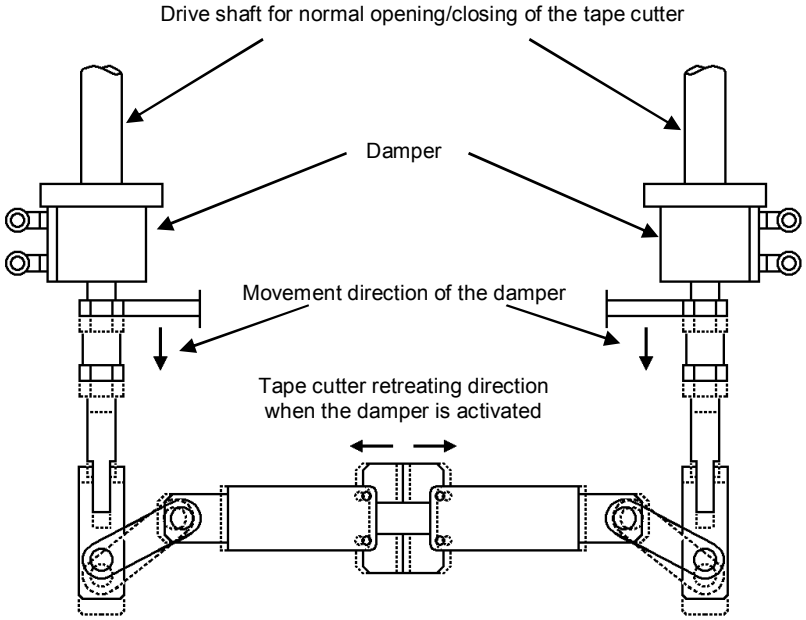


◆ Centering Function

- The centering function for correcting components off-centered on the tape prevents interference between components and the insertion pusher, thus achieving highly reliable insertion.

◆ Damper Type Tape Cutter Unit

- The tape cutters provided with a damper of 2.5 mm stroke (5 mm in all) reduce possible damage to the cutters in case foreign matter is caught in them.



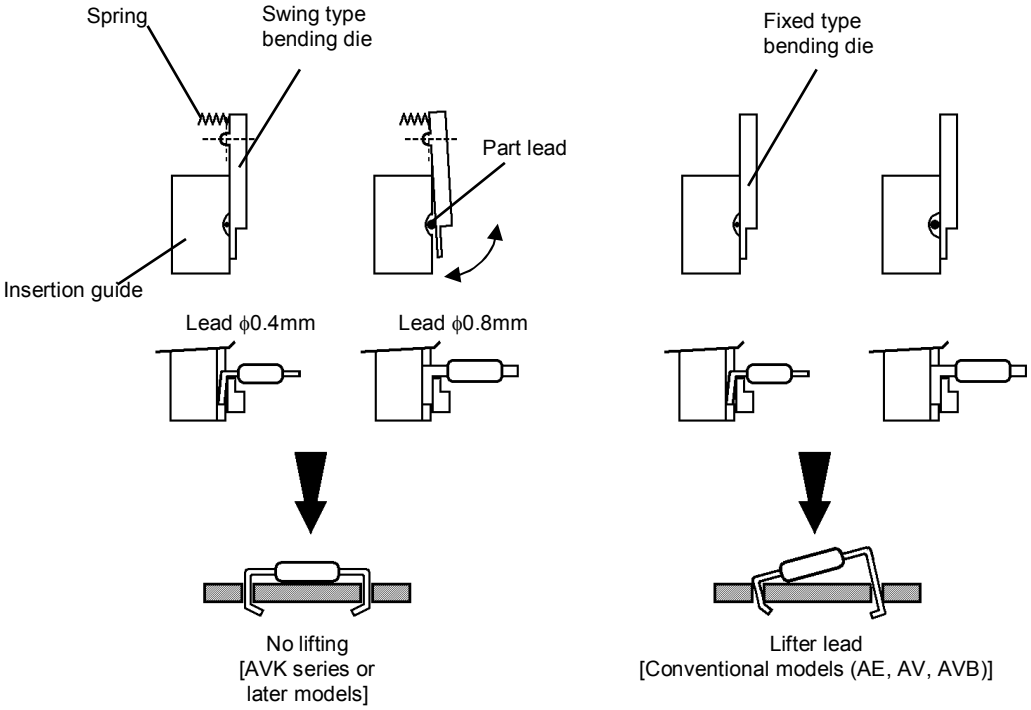
◆ Whole Area Self-correction Function as Standard Feature

- Recognizes both of two holes at the insertion position to make optimum positional correction for stable insertion.
  In addition, corrections of rotational error and insertion pitch are also enabled for highly accurate positioning (XY, θ and W), offering higher insertion rate.

■ Flexibilities

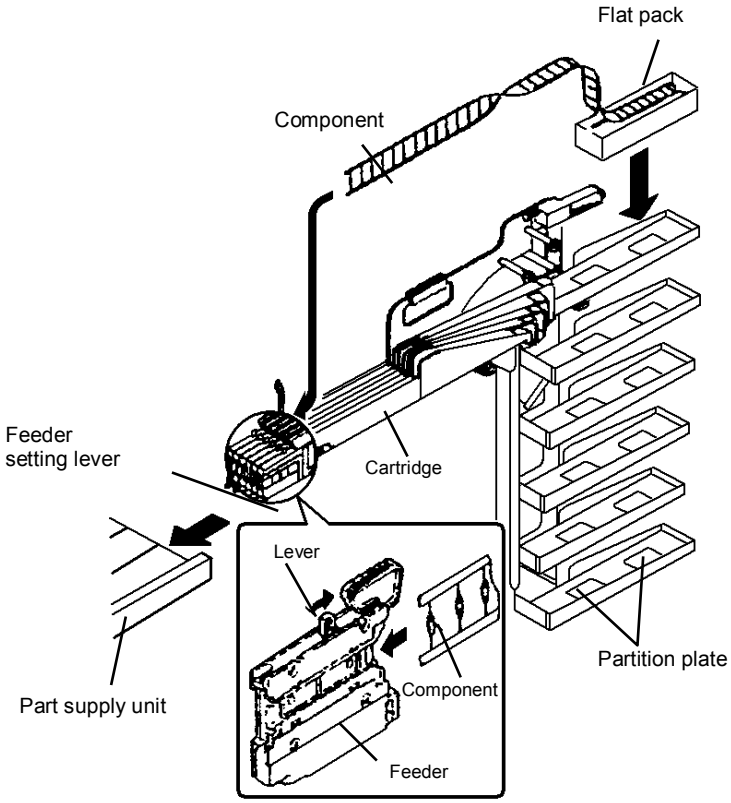
◆ Standardized Insertion Unit

- The swing type bending die allows a single tool to handle lead diameters of 0.4 to 0.8 mm and insertion pitches of 5 to 26 mm, improving flexibility.
- Switching of insertion unit according to the component is no longer required, improving the productivity significantly.



◆ Mixed Loading of 26 / 52 mm Feeders

- Adoption of 6-input vertical parts cartridge permits mixed loading of 26 mm and 52 mm feeders.
  Up to 120 inputs for NM-EJA3A and up to 60 inputs for NM-EJA2A are available respectively.



■ Improved Operability and Maintainability

◆ Color LCD Touch Panel

A color LCD touch-sensitive display on the control panel provides user-friendly interface, and allows the language to be selected among from English, Japanese and English with a single touch of a button.

Hierarchical menu structure

Operation screens classified in 3 modes (operator, engineer and service) and menu configuration simplified to the minimum needed level minimizes possible misoperations.

Tree jump

Enables you to go back to the main menu screen from any place within the submenu, offering improved operability.

Maintenance display

Required check items (daily, weekly, and monthly checks, etc) are indicated according to the cumulative operating hours. The date and time the check was made are saved.

◆ Adoption of Global Transformer

- Global transformer applicable to 200, 220, 380, 400, 420 and 480 V is built in. No external transformer is necessary.

◆ NC Data Compatibility

- The NC data from our conventional models (AV, AVB, AVF, AVG, AVP, AVK, AVK2, AVK2B and AVK3) can be used directly for the AV131.

◆ Compatibility of the Recommended Repair Parts

- The recommended repair parts used for the conventional models (AVK series) can be used for the AV131. (Note: Except for the lead cutter on the fixed side)

2.2. Mechanical Specifications

D79REC-12-020-A0

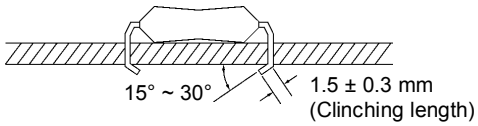
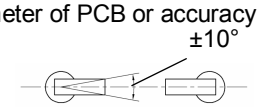
2.2.1 Machine Specifications

Power supply	● Rated voltage 3-phase 200 / 220 / 380 / 400 / 420 / 480 V selectable ● Frequency 50 / 60Hz ● Rated capacity Approx. 4kVA
Air supply	● Service air pressure 0.5 MPa ● Air consumption Approx. 150 L/min. (A.N.R) Note: Quick coupler attached (SMC KK13P-03M, KK13S-03F).
Dimensions	● NM-EJA3A W4070 × D1910 × H1610 mm (Not including signal tower) ● NM-EJA2A W2810 × D1910 × H1610 mm (Not including signal tower)
Mass	● NM-EJA3A Approx. 2,520 kg (main body only) ● NM-EJA2A Approx. 1,750 kg (main body only)
Environmental requirements	● Ambient temperature 20°C ± 10°C ● Humidity 25 ~ 75%RH (with no condensation) ● Height above sea level 1,000 m or below
Transport and storage	● Ambient temperature -20 ~ 60°C ● Humidity 75%RH or less (with no condensation)
Operating controls	● Interactive operation via color LCD touch screen (Front and Rear: Standard equipment) English/Japanese/Chinese selectable with a single button Recognition screen (displays the recognized board image on a superimposed window) Hierarchical menu (Operator/Engineer) ● I/O functions 3.5-inch FDD and SDD (standard equipment) ● Communication RS-232C
Coating color	● Standard color White: W-13 (G50)
Insertion tact time	● Max. tact time: 0.12 s/point Max. tact time conditions by component type and insertion direction

Insertion direction \ Applicable component	Axial lead component	Jumper wire
X (0°)	0.15 s/point	0.12 s/point
Y (90°)	0.12 s/point	0.15 s/point
-X (180°)	0.15 s/point	—
-Y (270°)	0.24 s/point	—

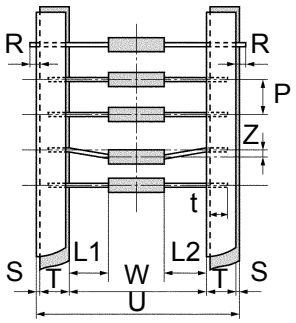
Other conditions for max. tact time (0.12 s/point)
● X-Y table movement
Within 35 mm for X and 39.5 mm for Y (When interlock 2 is selected)
(However, within 30 mm for X and 34 mm for Y when there are components above 3 mm high on the PCB. (When interlock 1 is selected))
● Feeder movement
None
(When the feeder moves by 1 pitch, the tact time will be 0.14 s/point.)

Pitch changeover	● 5.3 mm within the tact time (0.12 s/point)
Applicable components	● Taped axial lead components (carbon resistor, cylindrical ceramic capacitor, solid resistor, diode and taped jumper wire) ● Jumper wire (optional) Refer to 'Insertion Component Specifications' in this section for component dimensions.
Component packaging	● Flat pack

PCB replacement time	● Approx. 2 s (Ambient temperature: 20°C)
Applicable PCB	● Dimensions Min. 50 × 50 mm to Max. 508 × 381 mm Width-to-length ratio: Width should be equal to or less than length. Consult us for boards having width longer than length. ● Allowable insertion area Min. 50 × 40 mm to Max. 508 × 371 mm ● Thickness 1.6 ± 0.15 mm (for standard specification) *1 Although boards of thicknesses from 0.6 mm (min.) to 2.0 mm (max.) are applicable, changes to the transfer rail (option) are required if outside the above standard range. *2 Even if the board thickness is within the above range, clinched angle of the inserted lead or lead length may vary, or board transfer error may occur depending on the shapes of board or insertion holes. So it is recommended to perform an advance evaluation. ● Mass Max. 1 kg per board including inserted components. ● Flow direction Right to left (Standard) or Left to right (option) ● Board positioning method Pin positioning (Standard, without positioning in the Y direction), Edge positioning (Option, with positioning in the Y direction) *Contact us for the board positioning in the Y direction.
Insertion pitch	● 5 ~ 26 mm (can be set in 0.01 mm increments) (Jumper wire: 5 to 26 mm (Option))
Insertion direction	● 4 directions (X, Y, -X and -Y)
Anvil	● Cutting & Clinch type system (Piezoelectric detection system
No. of component inputs	NM-EJA3A 1. Taping width: 26 mm 120 inputs (60 + 60) + drum type jumper wire (Option) 2. Taping width 52mm (Pack width Max. 80) 120 inputs (60 + 60) + drum type jumper wire (Option) 3. Mixed loading of 26 mm and 52 mm tapes is available. NM-EJA2A 1. Taping width 26 mm 60 inputs + drum type jumper wire (Option) 2. Taping width 52 mm (Pack width Max. 80) 60 inputs + drum type jumper wire (Option) 3. Mixed loading of 26 mm and 52 mm tapes is available.
Insertion height	● Insertion height programmable according to the component diameter and lead diameter.
Clinching angle	● 15°to 30° inner clinching
Clinching length	● 1.5 ± 0.3 mm (Reference size: 1.5 mm)*  *The above numerical range indicates the variation which arises due to differences in the diameter or material of inserted component leads and in the diameter of holes of PCBs. *When changing the clinching length (changing reference size), the anvil unit parts need to be replaced. (Individually supported) **=REMARKS=** 1. A clinch mark will be left after cutting or clinching. The shape of clinch mark varies with the lead diameter or material of the inserted component. 2. Clinched component will be tilted a maximum of ±10° in the horizontal direction due to the hole diameter of PCB or accuracy of hole position. 

2.2.2 Insertion Component Specifications

■ Taped Axial Lead Component Specifications



Accuracy	I	II	III	Remarks
	$26^{+0.2}_{0}$	$26^{+0.5}_{0}$	$26^{+1.5}_{0}$	
W	$\left(52^{+0.2}_{0}\right)$	$\left(52^{+0.5}_{0}\right)$	$\left(52^{+1.5}_{0}\right)$	**=REMARKS= 4, 5, 6, 7, 8**
P	5 ± 0.3		* 5 ± 0.5	Tolerance for cumulative pitch error: ±2 mm/20 pitch
L1-L2	Max. 0.2	Max.0.5	Max. 1	
T	* 6 ± 1			
Z	Max. 1.0		* Max. 1.2	
R	0			* Lead wire terminals must not protrude from tape.
t	*Min. 3.2			Length of lead wire terminal on the section positioned on tape
S	*Max. 0.8			
U	Max. 65.5			
Remarks	5 mm pitch insertion			

=REMARKS=

1. Values with (*) conform to JIS standards. Others conform to Panasonic specifications.
2. When using sequence tape, make sure it conforms to the above requirements.
3. Use tin-plated lead wire. (Tin content: 95% or more)
4. Separate arrangements are required when inserting components with 5 mm pitch from 52 mm tape.
5. 52 mm taping with minus tolerance for W dimension require 52 mm special (for minus tolerance) feeder ($51^{+2.5}_{0}$).
6. For 52 mm taping with plus tolerance for W dimension, if 52 mm special (for minus tolerance) feeder ($51^{+2.5}_{0}$) is used, it is very likely that components will be off-centered.
7. The centering chuck cannot be used for fragile components such as glass diodes.
8. With insertion pitches within the tolerance of component off-center, even fragile components can be inserted without centering. (when using 52 mm special (for minus tolerance) feeder ($51^{+2.5}_{0}$)).
9. The edge shape of component may have an effect on centering.
10. The length of a component body must be up to 15 mm. Consult us if the length exceeds 15 mm.
11. Dimension U (U=S+T+U+W+T+S) must be within 65.5 mm.

2.2.3 Applicable Insertion Components

■ Axial Lead Components

Insertion pitch: 2 mm taping 5.0 to 12.7 mm pitch
52 mm taping 5.0 to 26.0 mm pitch

=HINT=

For 5.0 mm pitch insertion, use tape having accuracy of (I) in the table in 2.2.2

Lead diameter (d): ϕ0.4 to ϕ0.8 mm (Must not exceed ϕ 0.8.)

Body diameter (Dd): Max. ϕ 4.4 mm (When using 1.6 mm thick PCB)

=REMARKS=

Use lead wire surface-treated by tin plating. (Tin content: 95% or more)

=HINT=

The component diameter is limited by the thickness of the PCB. Refer to the table below.)

PCB thickness (mm) \ Body dia.	ϕ1.7	ϕ2.0	ϕ2.3	ϕ2.6	ϕ2.9	ϕ3.2	ϕ3.5	ϕ3.8	ϕ4.1	ϕ4.4	ϕ4.7	ϕ5.0
1.0	○	○	○	○	○	○	○	○	○	○	○	○
1.2	○	○	○	○	○	○	○	○	○	○	○	○
1.4	○	○	○	○	○	○	○	○	○	○	○	△
1.6	○	○	○	○	○	○	○	○	○	○	△	△
1.8	○	○	○	○	○	○	○	○	○	△	△	×
2.0	○	○	○	○	○	○	○	○	△	△	×	×

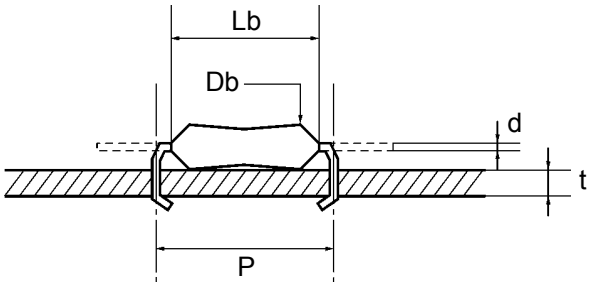
○…Insertion is enabled.
×…Insertion is disabled.
△…Contact us separately. (Shape of components (leads) inserted vary.)

Component length (Lb)

Taping accuracy (I): Max. Lb = Insertion pitch (P) - 1.6
Taping accuracy (II): Max. Lb = Insertion pitch (P) - 2.0
Taping accuracy (III): Max. Lb = Insertion pitch (P) - 2.7

=HINT=

In the case of 5 mm pitch insertion,
Max. Lb will be 3.4 mm (5-1.6 = 3.4)



=HINT=

- When using the centering chuck, components having a body length up to that of taping accuracy (I) can be inserted.
  (Checking with actual components is required because the centering accuracy differs with the shape of component edge.)
- Clinching radius varies with the diameter or material of lead wire.
  (Clinching radius will be smaller when the lead is thinner and softer, whereas it will be larger if the lead is thicker and harder.)
- Lead wires will have marks by the pusher in bending and by the clincher. Marks vary with the material of the leads of the inserted component.

■ Jumper Wire (Optional)

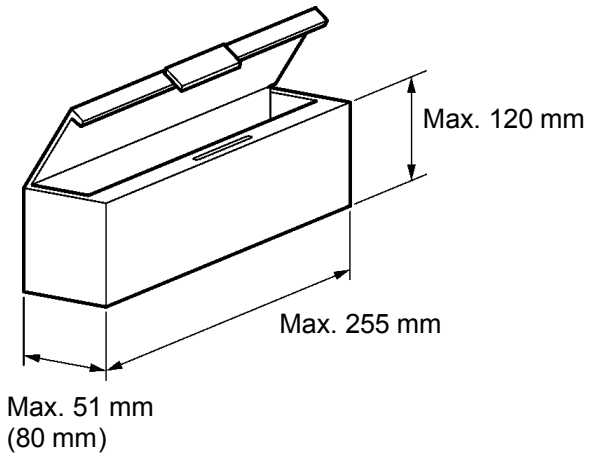
Insertion pitch (P): 5 to 26 mm

Wire diameter: ϕ0.6 ± 0.02 mm

* Use lead wire surface-treated by tin plating. (Tin content: 95% or more)

2.2.4 Packaging of Taped components

■ Flat Pack



=HINT=

The dimension in the parentheses applies to 52 mm taped components.

- If the width of the tape box is less than 80 mm, standard cartridges can be used.
- If the width of the tape box is less than 85 mm, use cartridges for 85 mm.
  Consult with us separately if the tape box width exceeds 85 mm.
- The maximum allowable mass of a flat pack is 1 kg. The total mass of the flat packs per cartridge must be up to 4.5 kg.

2.3. Control Specifications

D79REC-12-030-A0

2.3.1 Control Specifications

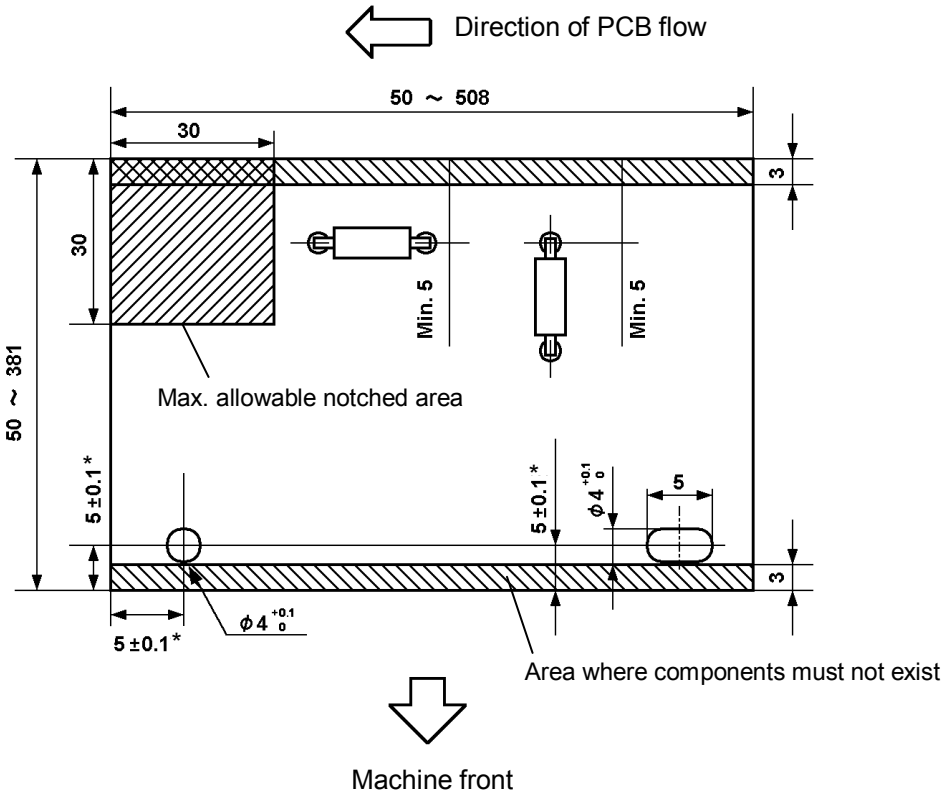
Control system	● Microcomputer system
Programming method	● X-Y axes: Absolute coordinate commands (Incremental coordinates can be displayed when editing recognition data.) ● Insertion pitch Absolute commands ● Z axis (Component feeder) Feeder number designation
Least command increment of X-Y table	● 0.01 mm / pulse
Program	Number of NC programs ● Built-in flash memory 200 programs 5000 steps / program (Note: In case of programs involving pattern repeat, the total number of steps after expanding the program must be less than 5,000.) ● SD card Depends on the card capacity. (up to 512 MB)
Indication light	● Signal tower (3 lamps, top to bottom: red/yellow/green or yellow/white/green)
Hole position offset	● Determines optimum insertion positions based on the recognition of all insertion positions to make positional corrections (XY, θ and pitch).
Self-diagnosis function	● Presents on the color LCD the machine status when the power is ON and situation of any troubles occurred, as well as countermeasures against the troubles.
Others	● Programming functions Refer to 'Programming Manual' for details. ● PanaPROJ/CAM PanaPROJ/CAM, the software for optimizing the balance of line consisting of multiple machines and for assigning data is available. ● Data compatibility Data from conventional AV series (AV, AVB, AVF, AVG, AVP, AVK, AVK2, AVK2B and AVK3) can be directly used on the AV131.

*When restarting the machine after a long stop, check that there is nothing wrong with the machine operations in Manual mode and idle the machine.

2.4. PCB Design Standard

D79REC-12-040-A0

2.4.1 PCB Specifications

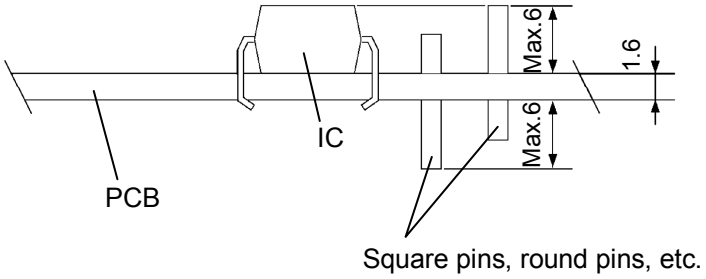


=HINT=

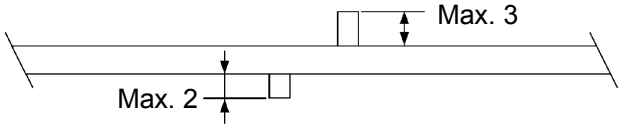
- The dimensions of 5 mm marked with an asterisk (*) (from center of hole to edge of board) are standard values.
- Detailed arrangements are required when using PC boards having notches or cutouts.

2.4.2 Restrictions on Height of Components Already Mounted on Top and Bottom Sides of PCB

(i) When X-Y table movement within tact time is X = 30 mm, Y = 34 mm

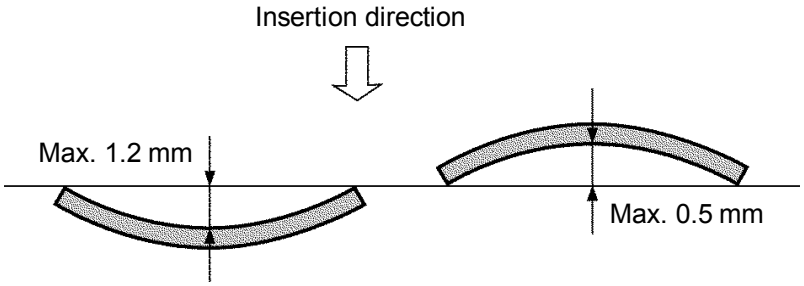


(ii) When X-Y table movement within tact time is X = 35 mm, Y = 39.5 mm

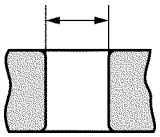
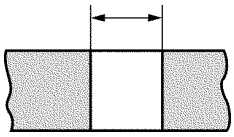


* Can be switched through on-screen setting.

2.4.3 PCB Accuracies

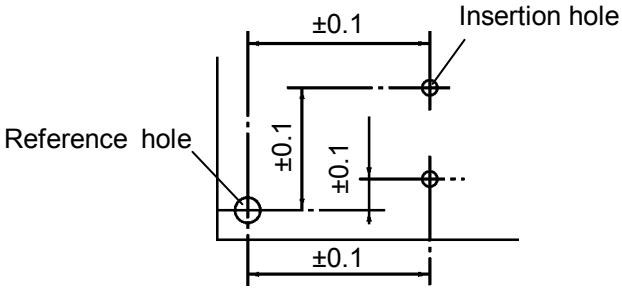
Item	
Board thickness	1.6 ± 0.15 mm, Option: 0.6 to 2.0 mm (set in 0.2 mm increments) Consult us separately for thickness other than the above.
Allowable board warp	

Insertion hole diameters

	Hole type and diameter	
	(Punched hole)	(Drilled hole)
Lead diameter		
$\phi 0.8^{0}_{-0.05}$	$\phi 1.2^{+0.1}_{0}$	$\phi 1.3^{+0.1}_{0}$
$\phi 0.6 \pm 0.05$	$\phi 1.0^{+0.1}_{0}$	$\phi 1.1^{+0.1}_{0}$
$\phi 0.5 \pm 0.05$	$\phi 0.9^{+0.1}_{0}$	$\phi 1.0^{+0.1}_{0}$
$\phi 0.4^{+0.05}_{0}$	$\phi 0.8^{+0.1}_{0}$	$\phi 0.9^{+0.1}_{0}$

=REMARKS=
Use of hole diameters other than those listed above may affect the insertion rate or the securing of inserted components.

Insertion hole position accuracy



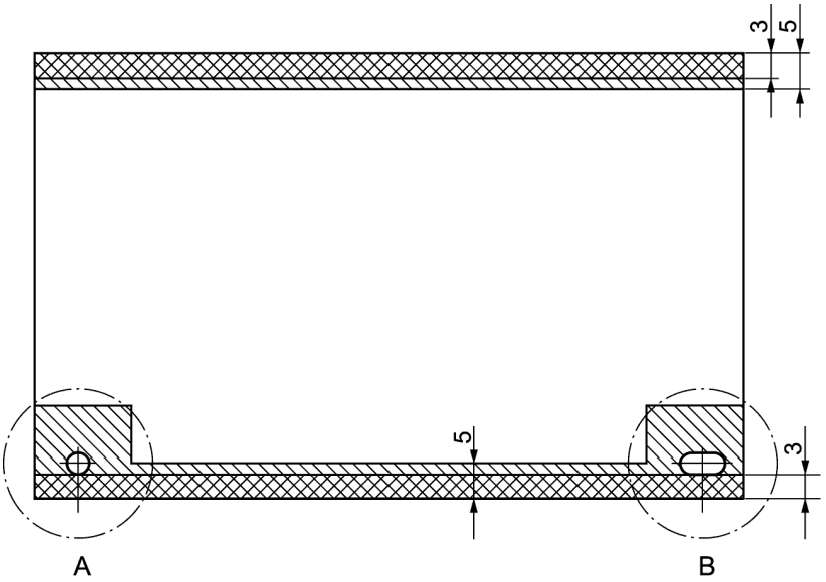
Variation among PCBs must be within ± 0.1 mm.

2.5. Insertion Restrictions

D79REC-12-150-A0

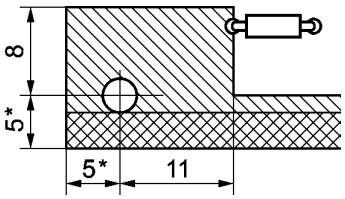
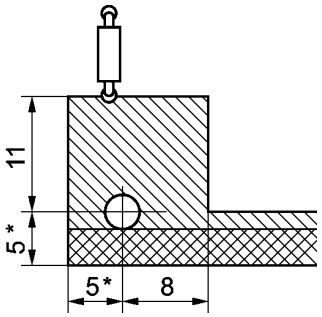
2.5.1 Insertion Restrictions

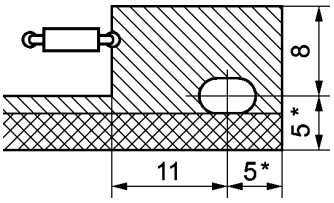
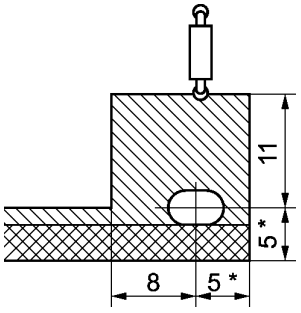
■ Dead Space around Reference Holes



: Area where component must not exist

: Dead space

Detailed dimensions of portion A	
Dead space for insertion in the 0° and 180° directions (Insertion impossible) (X, -X)	Dead space for insertion in the 90° and 270° directions (Insertion impossible) (Y, -Y)
	

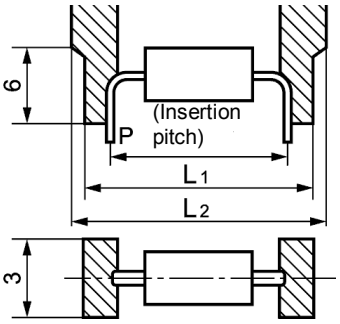
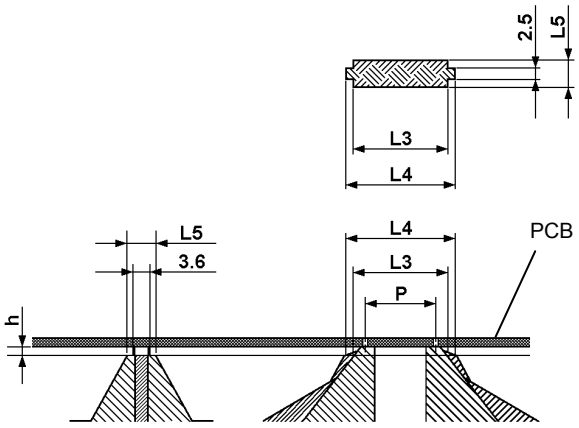
Detailed dimensions of portion B	
Dead space for insertion in the 0° and 180° directions (Insertion impossible) (X, -X)	Dead space for insertion in the 90° and 270° directions (Insertion impossible) (Y, -Y)
	

*Dimensions marked with an asterisk (*) are for standard specification.

■ Dead Space due to Insertion Tooling

The insertion tooling contacts the PCB during insertion. That is, the shaping mechanism contacts the top of the PCB, and the anvil contacts the bottom in the shaded areas in the figures.

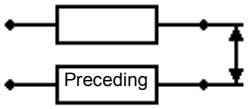
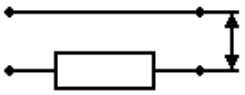
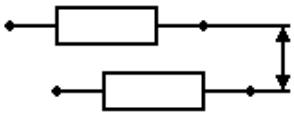
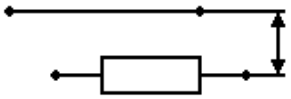
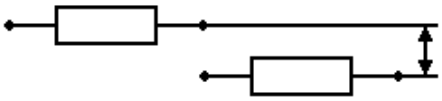
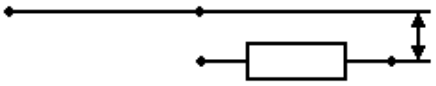
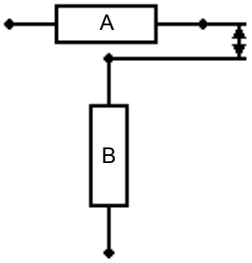
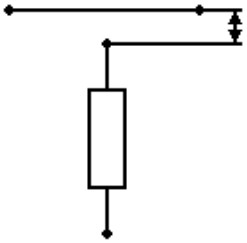
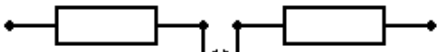
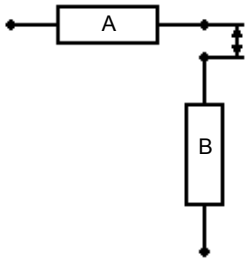
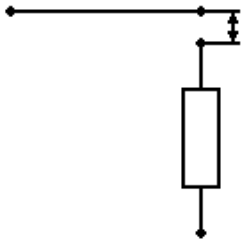
This dead space must be allowed between the component currently being inserted, and the adjacent previously inserted components.

Dead space due to bending base	Dead space due to anvil
	

L_1	L_2	h	L_3	L_4	L_5
P + 2.0	P + 7.5	0	P + 2.2	P + 2.2	3.6
		1	P + 4	P + 4	3.6
		2	P + 5.6	P + 9	6
		6	P + 12	P + 14	11

■ Calculation of Clearance Required between Adjacent Parts

Provide an allowance of 0.2 mm for the values obtained by the following calculations.
Jumper wire diameter: d, Lead diameter of the previously inserted part: d1, Diameter of the previously inserted part: D1, Diameter of the part to insert: D2.
However, the center of the lead wire and that of the insertion hole must be identical.

Between axial parts	Between an axial part and jumper wires
 $\frac{3.6+d1}{2}$ or $\frac{D1+D2}{2}$	 $\frac{3.6+d}{2}$
 $\frac{3.6+D1}{2}$ or $\frac{3.6+d1}{2}$ (whichever is greater)	 $\frac{3.6+d}{2}$
 $\frac{3.6+d1}{2}$	 $\frac{3.6+d}{2}$
 To insert A first: $\frac{2+D1}{2}$ To insert B first: $\frac{3.6+d1}{2}$	 $\frac{2+d}{2}$
 $\frac{4+d1}{2}$	$\frac{4+d}{2}$
 To insert A first: $\frac{4+d1}{2}$ To insert B first: $\frac{3.6+d1}{2}$	 $\frac{4+d}{2}$

3. VARIOUS FUNCTIONS

This chapter describes the operation of various functions used for production.

3.1. Multi-Interlock Function

D79REC-33-200-A0

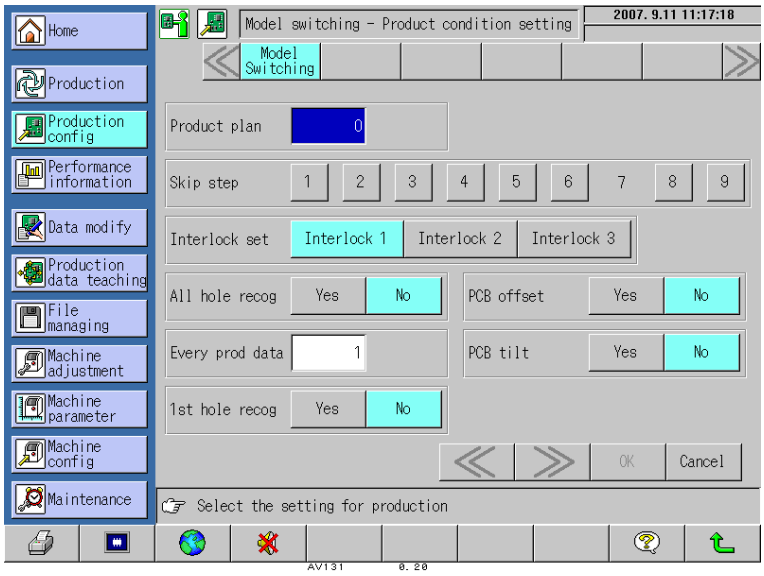
3.1.1 Multi-Interlock Function

This function enables the moving range of the X-Y table to be changed according to the height of components already placed on PC boards.
When there are no placed components or only short components have been placed on the PC board, setting 'Interlock 2' makes the movement of the X-Y table within the tact time longer than that when 'Interlock 1" is selected.
Usually, specify 'Interlock 1'.

=REMARKS=
Setting 'Interlock 2' inadvertently may cause the insertion head to contact and damage the already placed components.

The Multi-Interlock can be set under the Production config menu.

Production config
└ Model switching
└ Interlock set



=HINT=
When 'Interlock 3' setting, it is automatically switched to 'Interlock 2' when the jumper wire is inserted and switched to 'Interlock 1' when the axial lead component is inserted.

3.2. Recovery Data Compensation Function

D79REC-33-220-A0

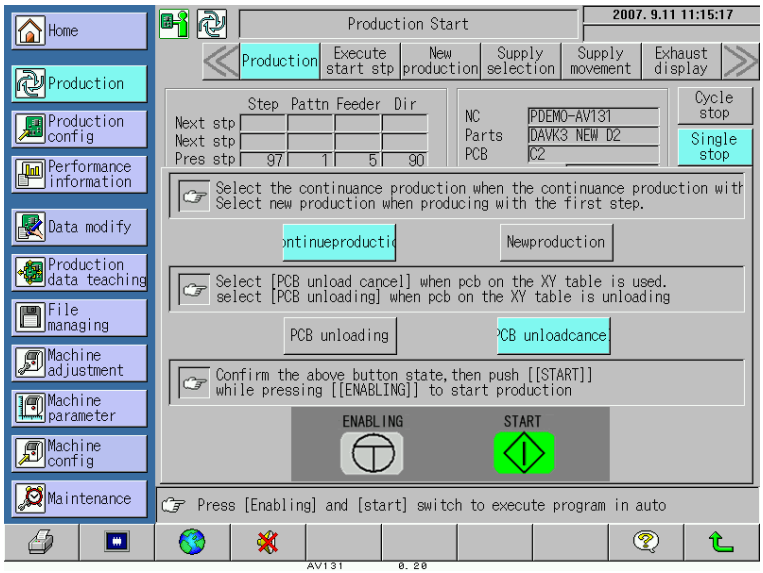
3.2.1 Recovery Data Compensation Function

Even if, after an error which disables continuous production has occurred, the error is reset and another function from the menu is attempted or power is turned OFF, this function allows the production to be resumed from the step where the error occurred, by pressing the [Continue production] followed by <START>.
This function is effective in resuming production in case the machine has been stopped (or power has been turned OFF) by mistake after an error disabling continuous production.

◆ Basic operation of the recovery data compensation function

1. An error occurs during production.
2. Reset the error. (Close the error screen or turn the power off.)
3. If the head is not at M.Origin, set the head servo free, and return the head to origin by rotating it forward.
4. If the head is holding a component, use the machine adjustment function to remove the component. (Pressing <START> without removing it will cause the two components to be inserted in one point.
5. Selecting the [Production] menu displays the Production start screen shown below. Press <Continue production> to turn ON the button. Then press <ENABLING> and <START> at the same time.

=HINT=
To cancel the recovery data compensation function:
- Press [New production] .



3.3. Backlight Setting

D79REC-33-630-A0

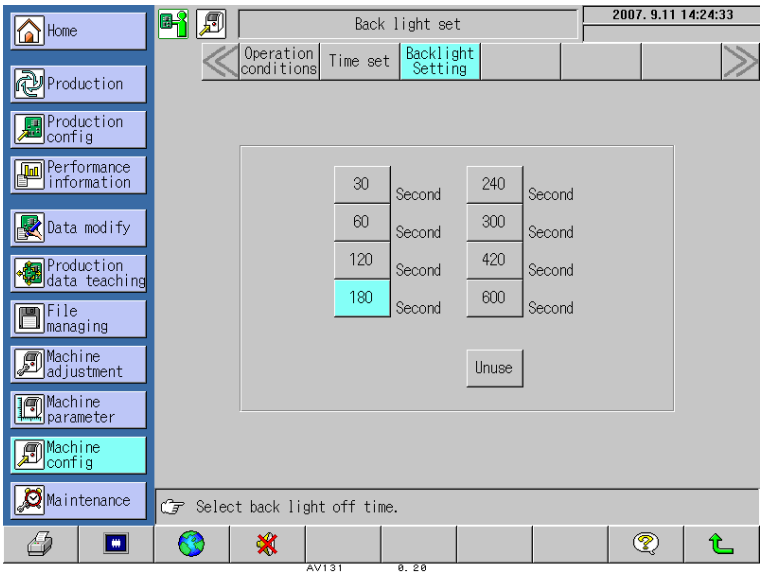
3.3.1 Backlight Setting

This function automatically turns OFF the backlight and blanks out the screen after the touch panel on the control panel has not been operated for a certain period of time.

The screen is restored if any button on the touch panel is pressed, an error message occurs, or if the PIR sensor on the control panel is activated. The time interval for screen saver can be set on the backlight set screen.

Machine config (Main menu)

└ Back light setting (Sub-menu)



- Set the interval before the screen saver is activated.
  Set any of the following: 30, 60, 120, 180, 240, 300, 420 or 600 seconds
  Pressing [Unuse] does not activate the screen saver function.

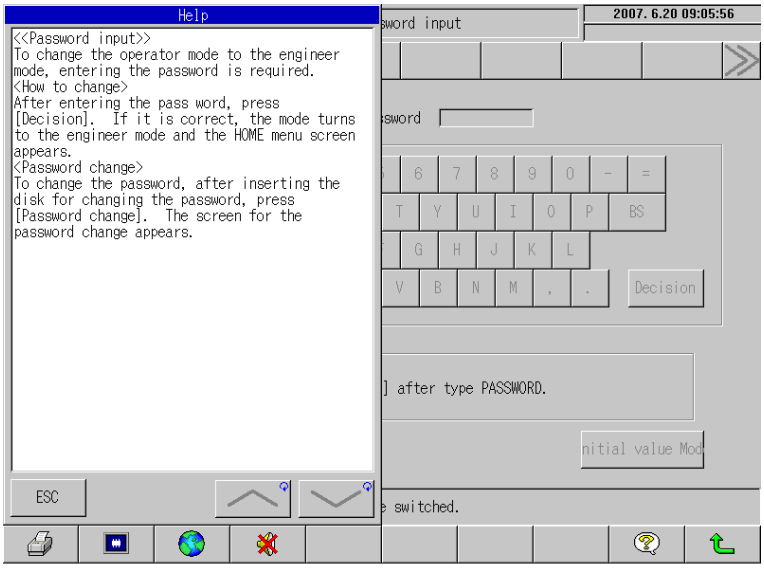
Press to determine this setting.

3.4. Help Information

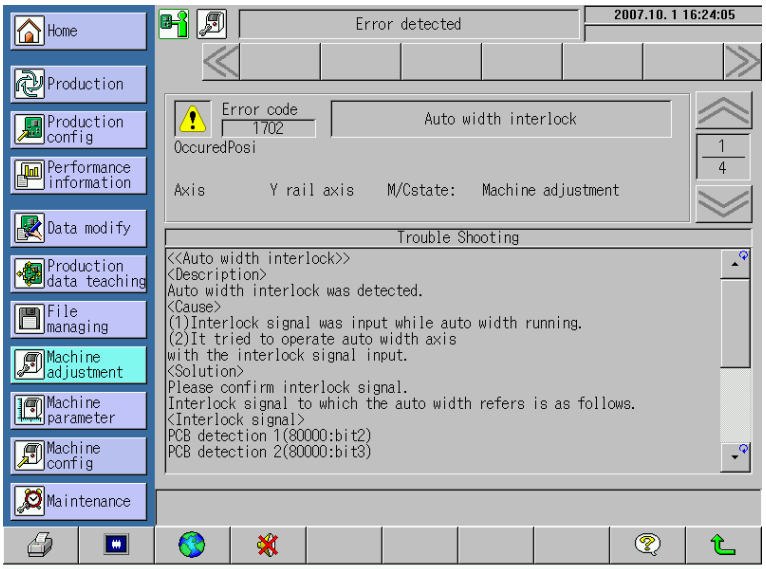
D79REC-33-620-A0

3.4.1 Help Information

Pressing at the bottom of each screen displays the help for the current screen.



When an error has occurred, detailed information is displayed as “Description”, “Cause” and “Solution” under Trouble Shooting.



4. SCREEN TRANSITION

This chapter provides the screen transition diagrams.

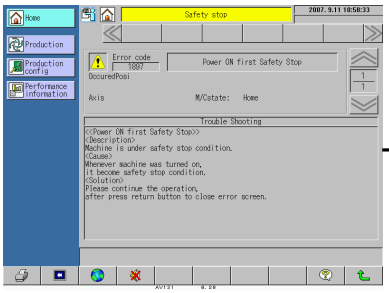
4.1. Screen Transition Diagrams

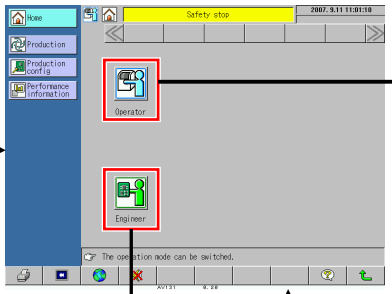
D79REC-21-010-A0

4.1.1 Home

This screen always appears first when power is turned ON.

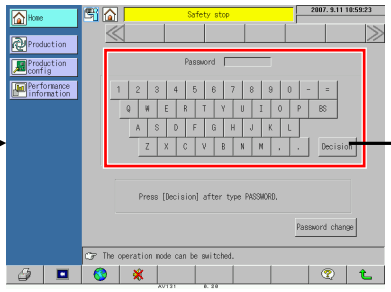
Home menu is displayed.

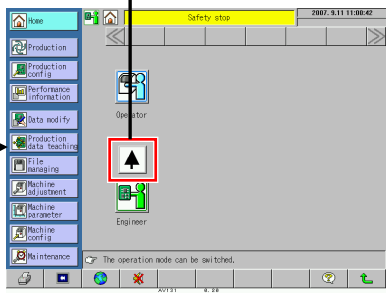




Pressing the Operator button does not change the screen.

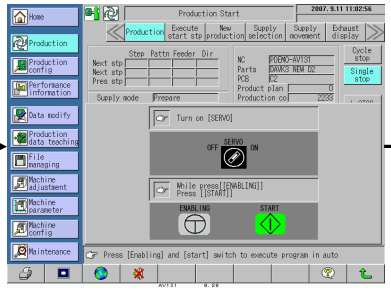
Pressing the arrow button returns the display to the Operator mode.



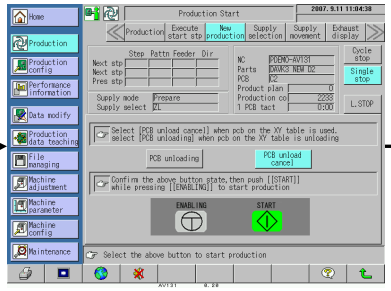


Inputting password followed by pressing [Decision] engages the Engineer mode.

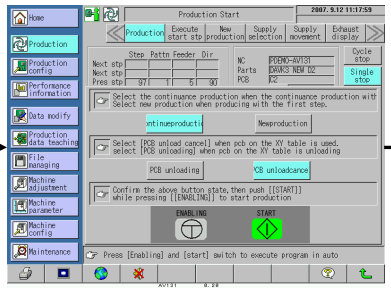
4.1.2 Production



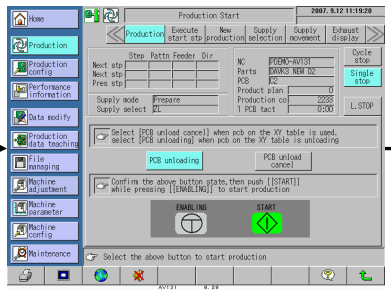
- When the [Production] menu is selected with no execution steps stored and there is no PC board on the XY table.



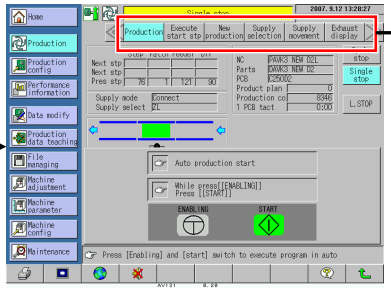
- When the [Production] menu is selected with no execution steps stored and there is a PC board on the XY table.

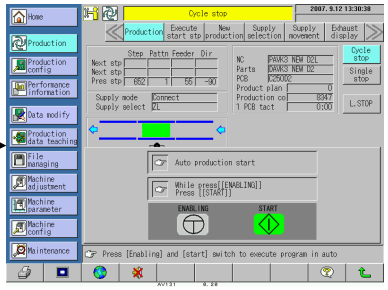


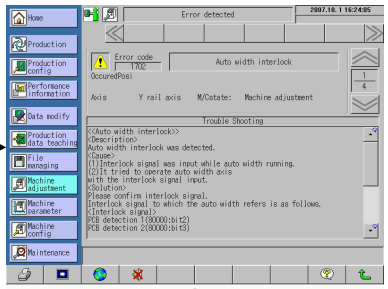
- When the [Production] menu is selected with execution steps stored and there is no PC board on the XY table.

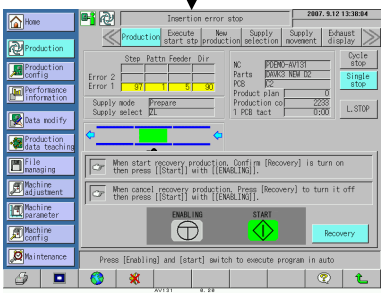


- When the [Production] menu is selected with execution steps stored and there is a PC board on the XY table.

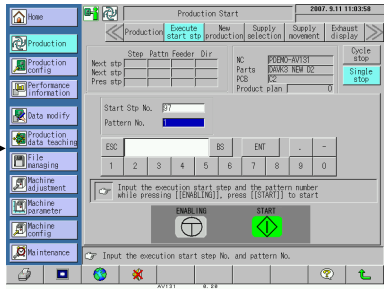




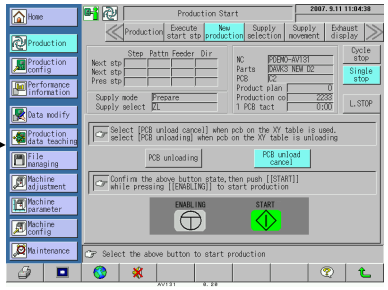




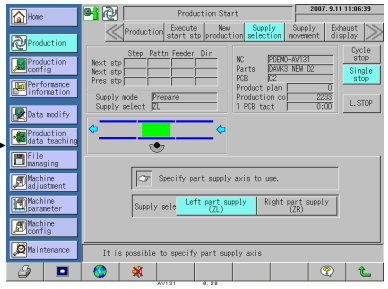
- Execute start step



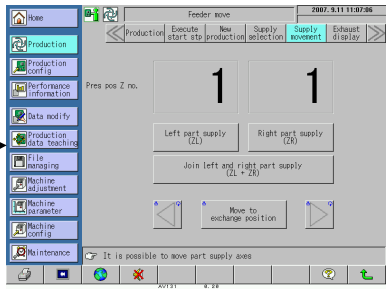
- New production



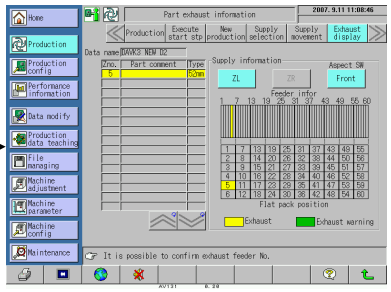
- Select Supply selection



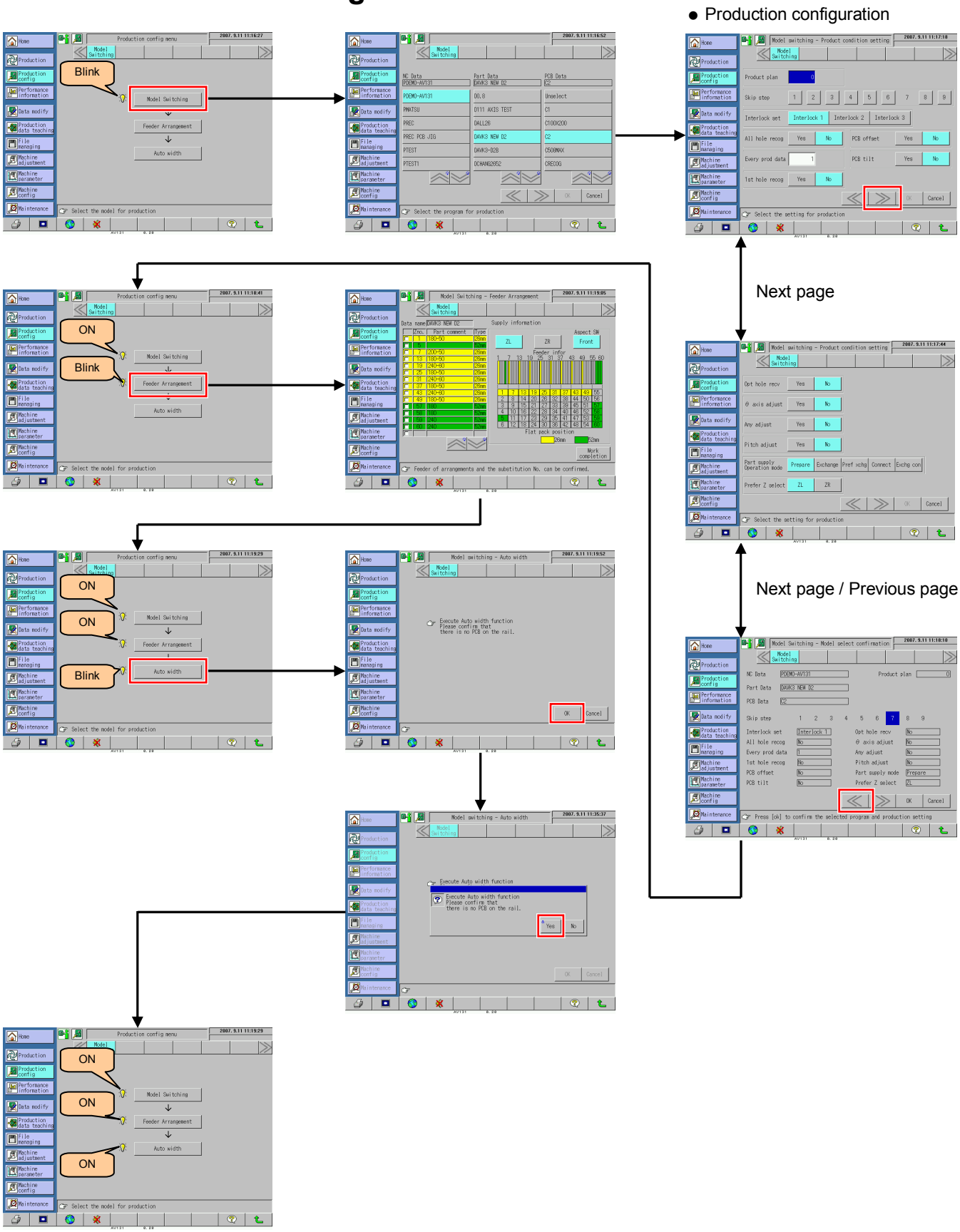
- Part supply movement



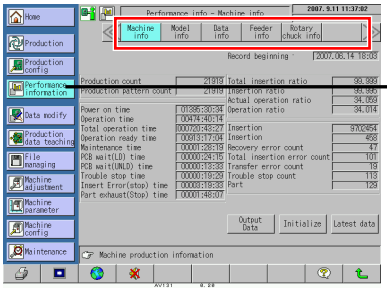
- Exhaust display



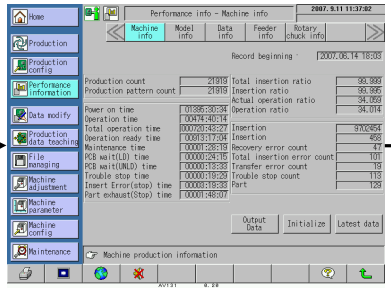
4.1.3 Production Configuration



4.1.4 Performance Information

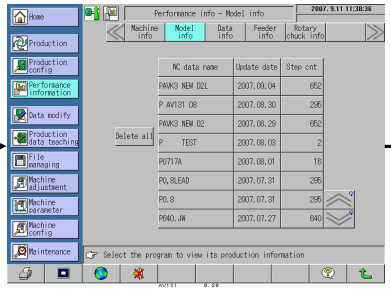


- Machine information



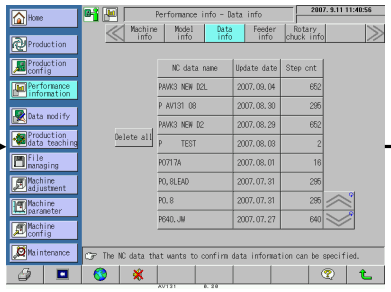
P 4.1-5

- Model information



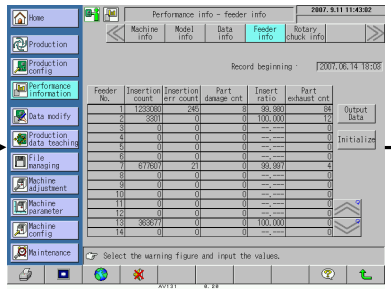
P 4.1-5

- Data information



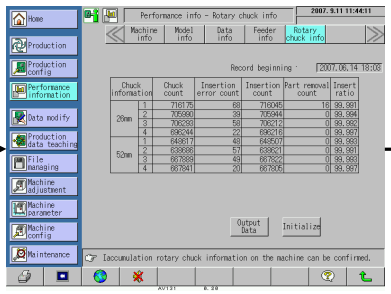
P 4.1-6

- Feeder information



P 4.1-6

- Rotary chuck information

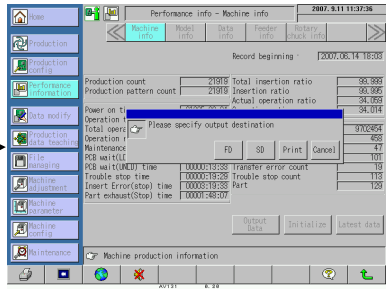


P 4.1-7

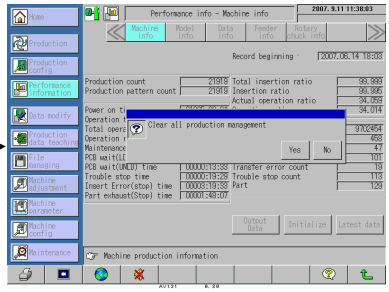
■ Machine Information

- "Machine information" screen

● Data output



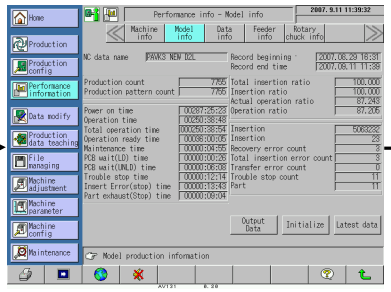
● Initialize



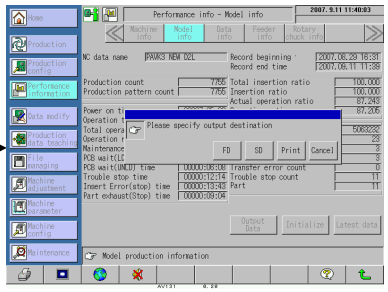
■ Model Information

- "Model information" screen

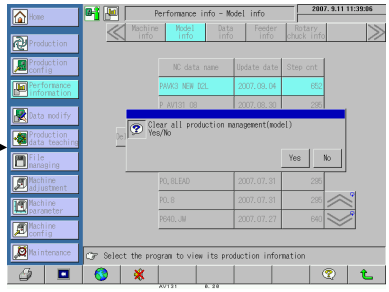
● Select data



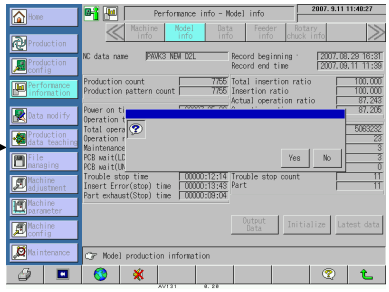
● Data output



● All clear



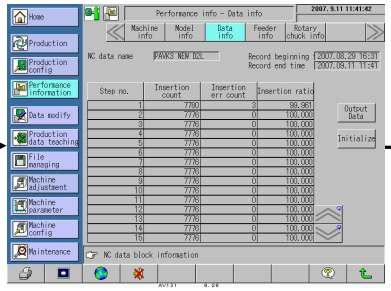
● Initialize



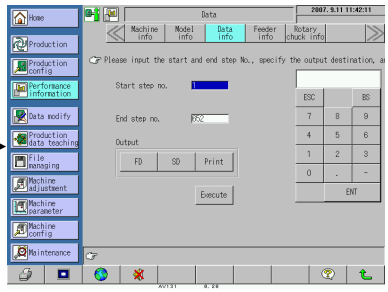
■ Data Information

- "Data Information" screen

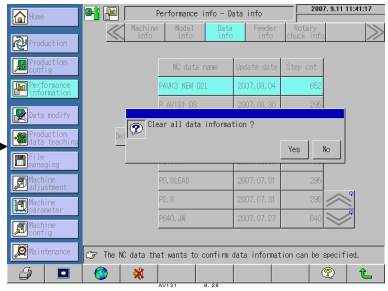
● Select data



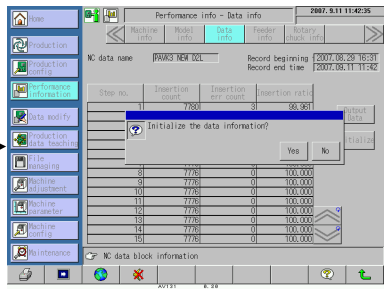
● Data output



● All clear



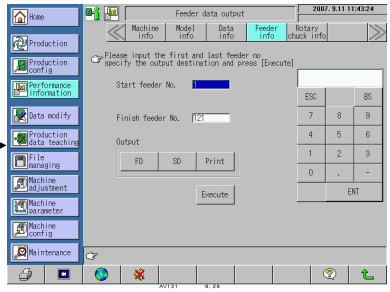
● Initialize



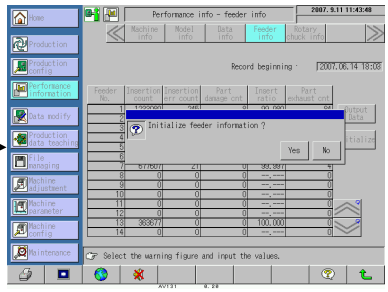
■ Feeder Information

- "Feeder Information" screen

● Data output



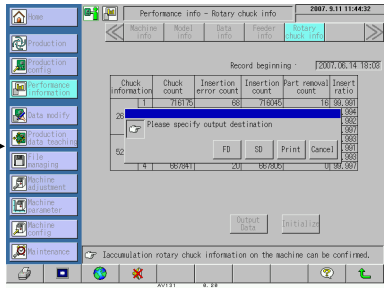
● Initialize



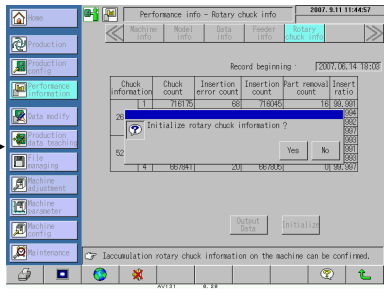
■ Rotary Chuck Information

- "Rotary Chuck Information" screen

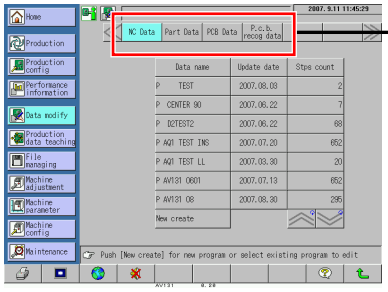
- Data output



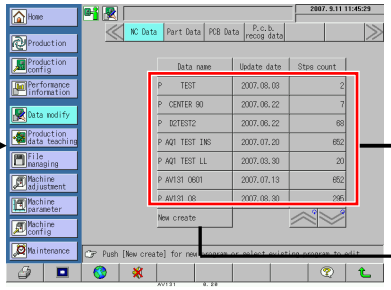
- Initialize



4.1.5 Data Modify



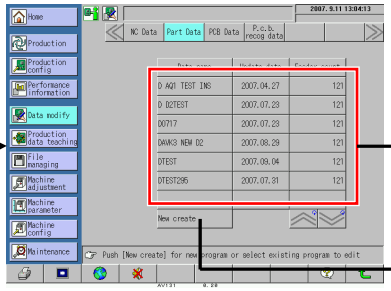
● NC data



P 4.1-9
◆ Edit NC data

P 4.1-9
◆ New create

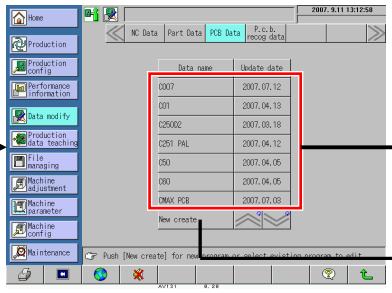
● Part Data



P 4.1-12
◆ Edit Part Data

P 4.1-12
◆ New create

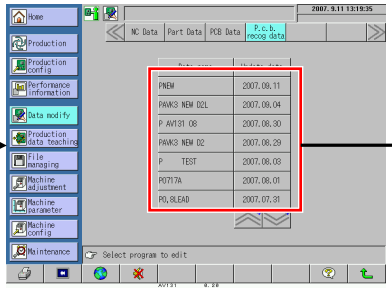
● PCB Data



P 4.1-13
◆ Edit PCB Data

P 4.1-13
◆ New create

● P.c.b. recognition data

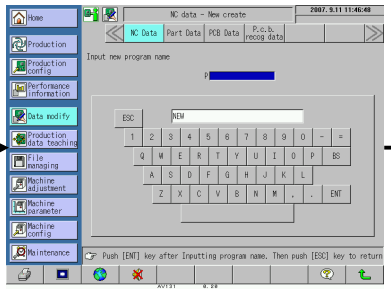


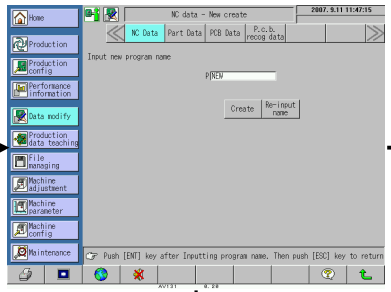
P 4.1-10
Select P.c.b. recog data
/ Edit

■ NC Data

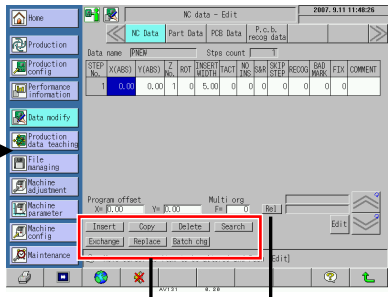
◆ New Create

● Input NC data name



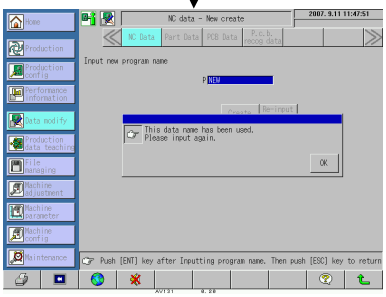


● Edit data



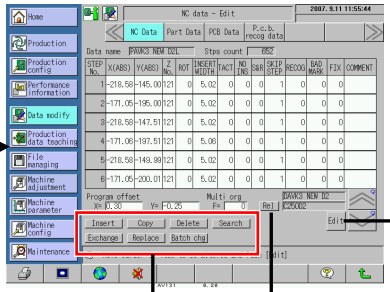
A — P 4.1-10

B — P 4.1-11



Error message for duplicate NC data name entry

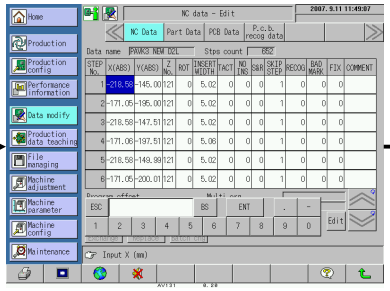
◆ Edit data



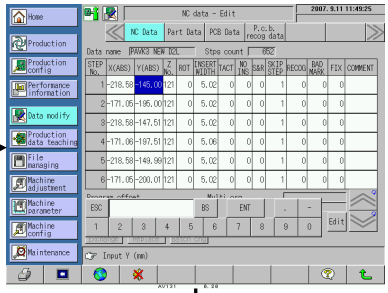
A — P 4.1-10

B — P 4.1-11

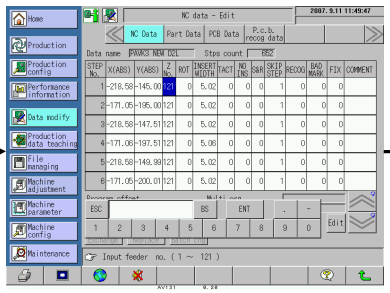
● X (ABS)



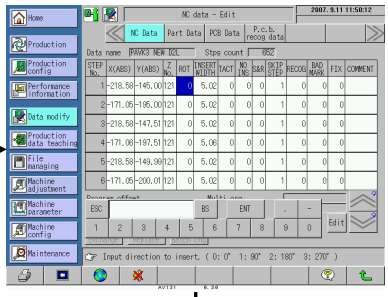
● Y (ABS)



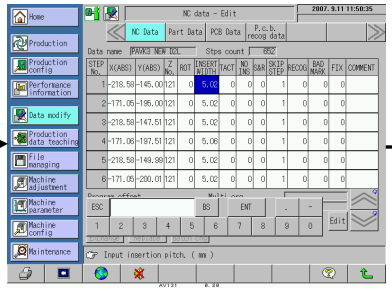
● Z No.



● ROT



● INSERT WIDTH



1 — P 4.1-10

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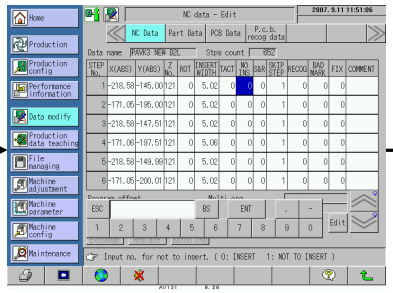
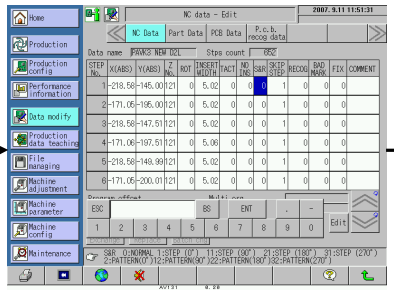
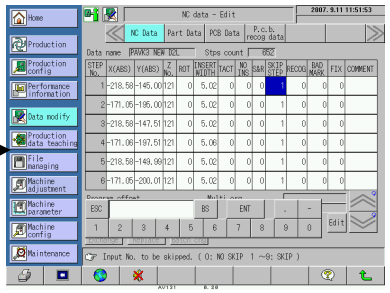
● NO INS



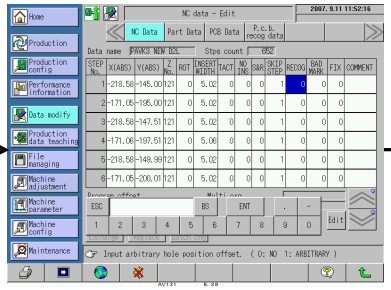
● S&R



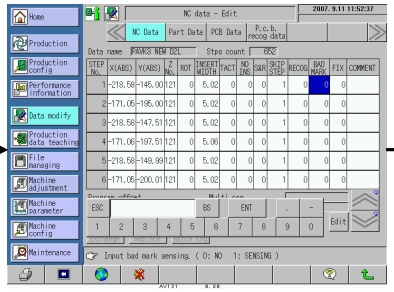
● SKIP STEP



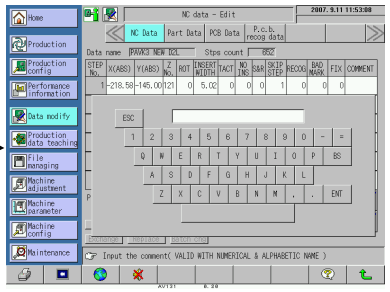
● RECOG



● BAD MARK



● COMMENT



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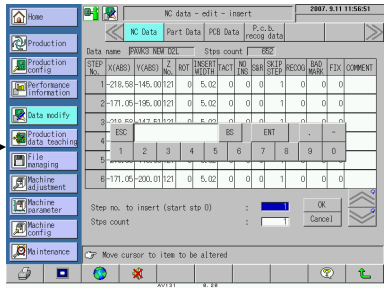
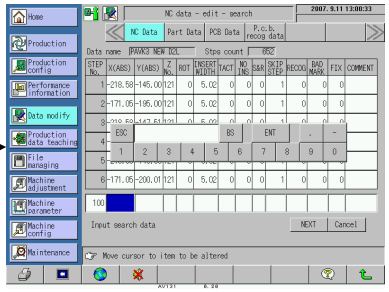
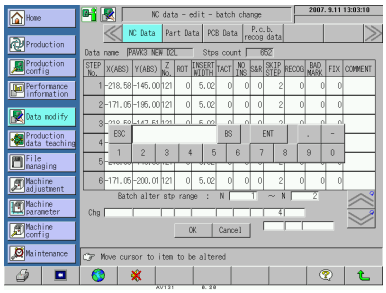
● Insertion



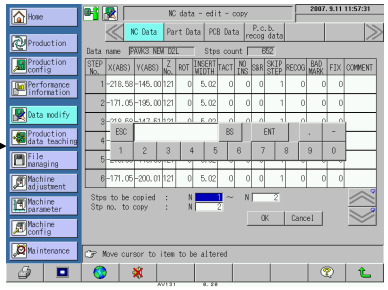
● Search



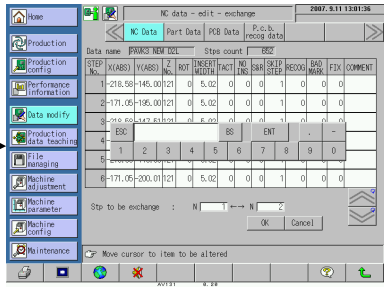
● Batch alter



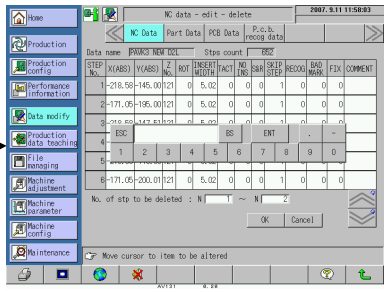
● Copy



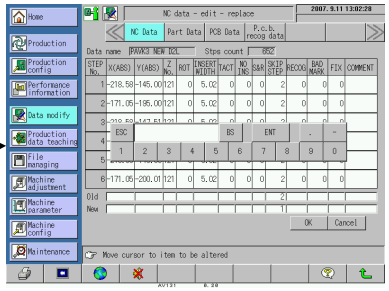
● Exchange



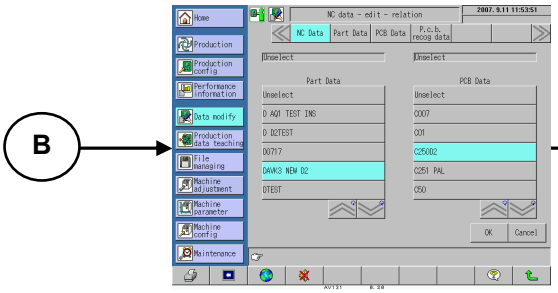
● Delete



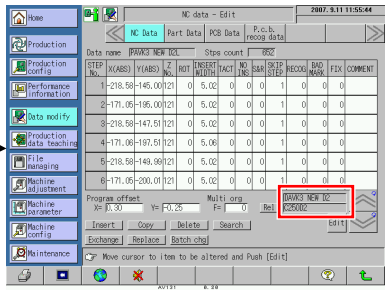
● Replace



● NC data – edit - relation



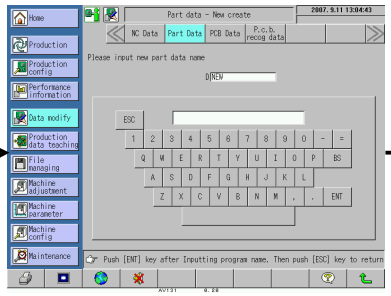
● Relation data

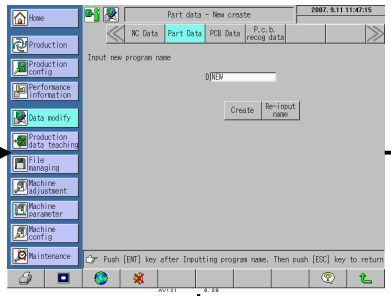


■ Part Data

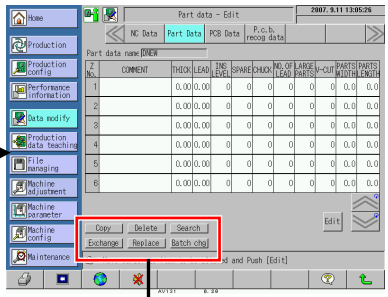
◆ New Create

● Input part data





● Edit data



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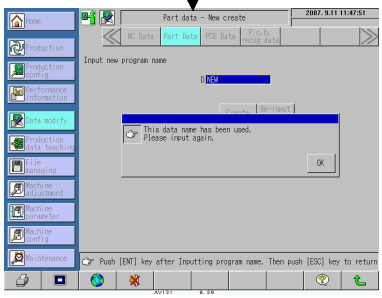
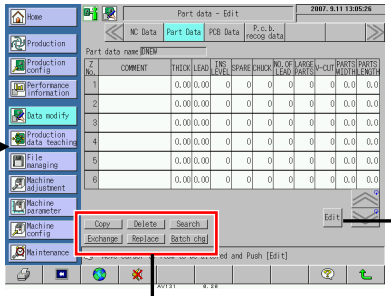
P 4.1-12



Error message for duplicate part data name entry

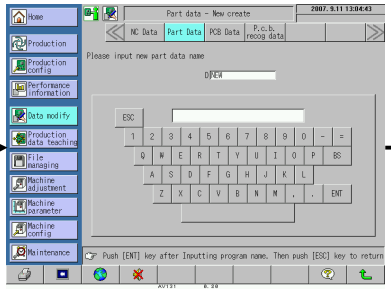
◆ Edit data



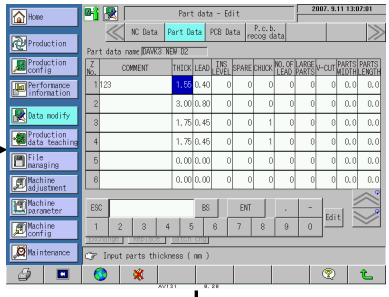
A

P 4.1-12

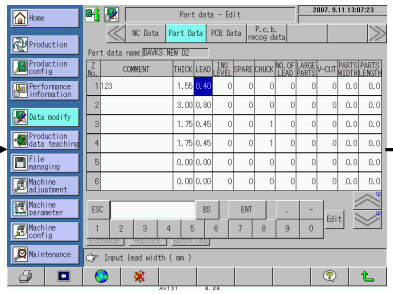
● COMMENT



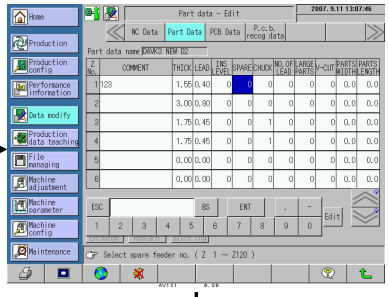
● THICK



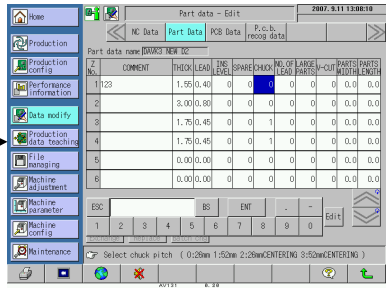
● LEAD



● SPARE



● CHUCK

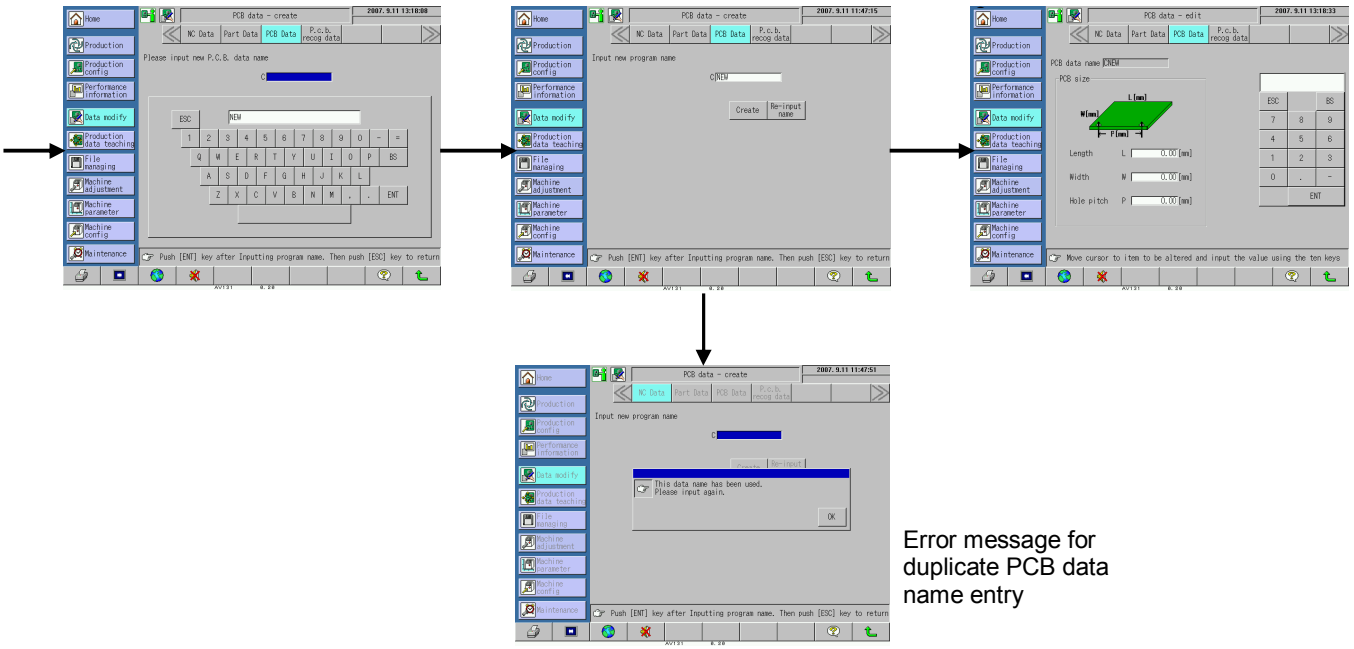


A
● Copy
● Search
● Replace
● Delete
● Exchange
● Batch change

■ PCB Data

◆ New Create

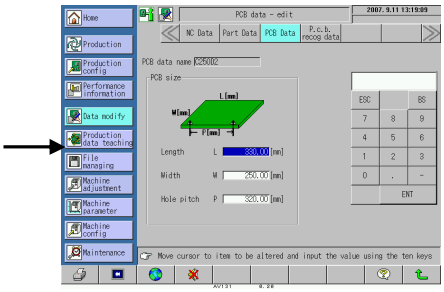
● Input PCB data name



● Edit data

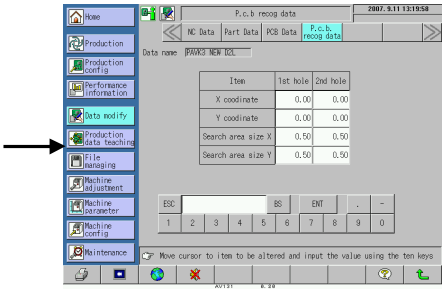
Error message for duplicate PCB data name entry

◆ Edit data



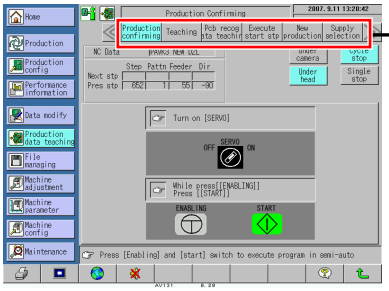
■ PCB Recognition Data

◆ Edit data

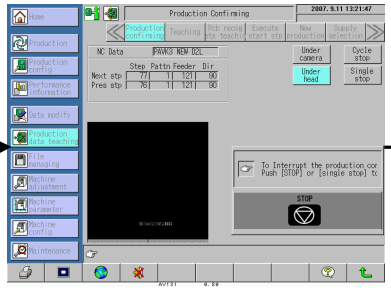


4.1.6 Production Data Teaching

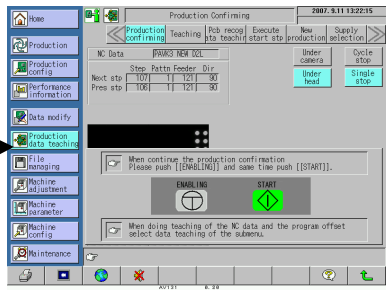
- Production data teaching



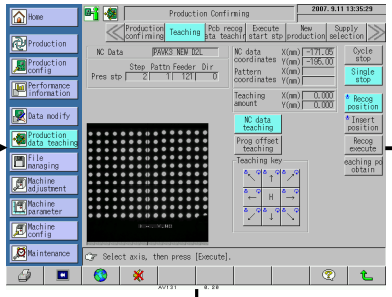
- Production confirming



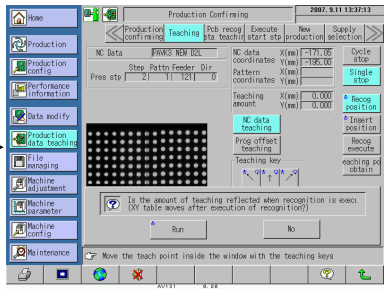
- Single stop



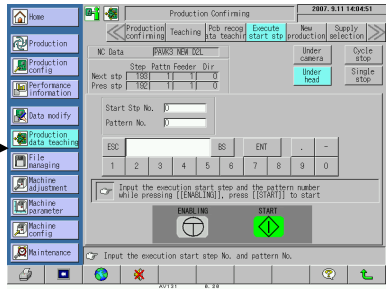
- NC data teaching



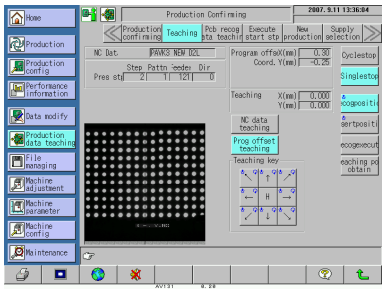
- Production confirming



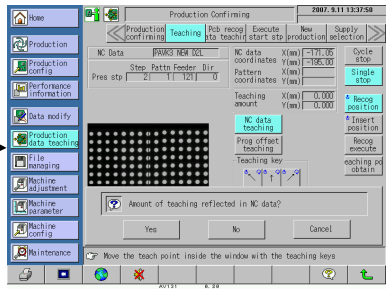
- Execute start step



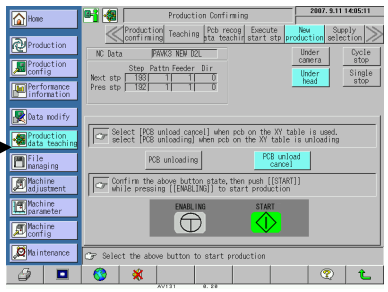
- Program offset teaching



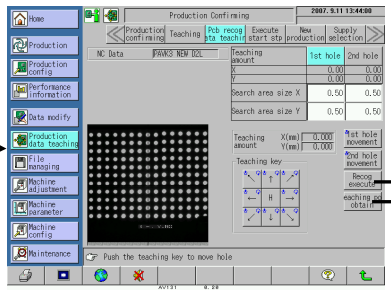
- Recognition position

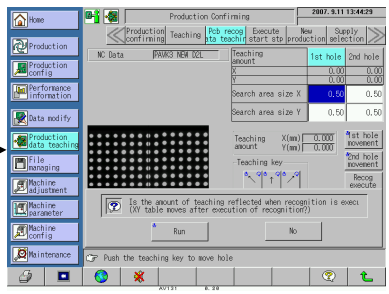


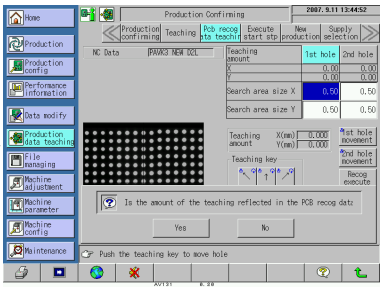
- New production



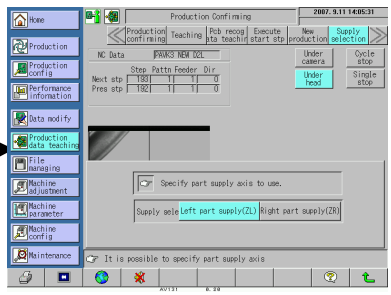
- Pcb recognition data teaching





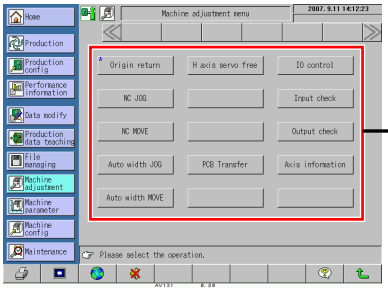


- Supply selection

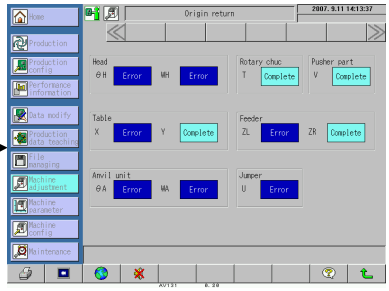


4.1.7 File Managing

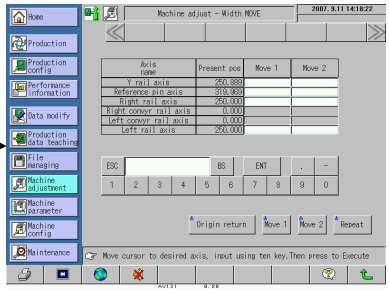
4.1.8 Machine Adjustment



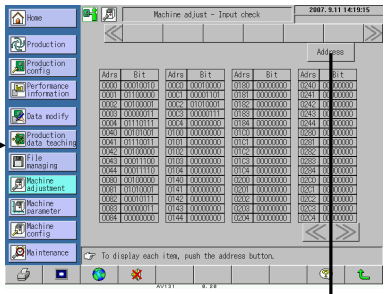
● Origin return



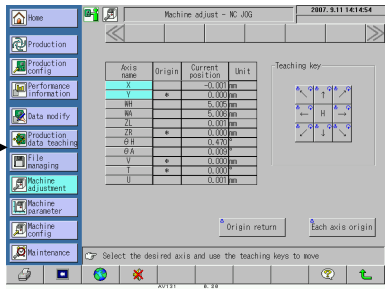
● Width MOVE



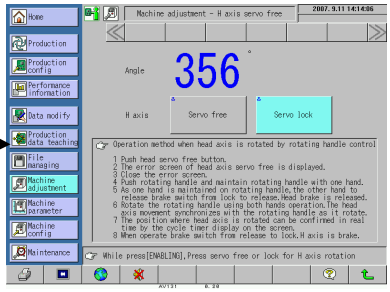
● Input check

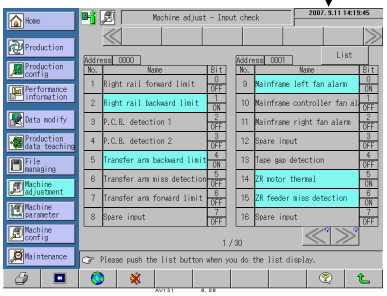


● NC JOG

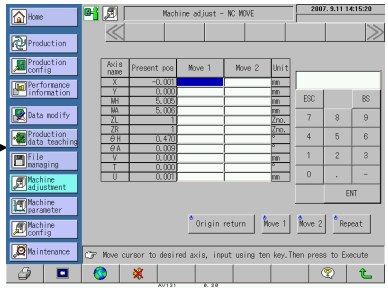


● H axis servo free

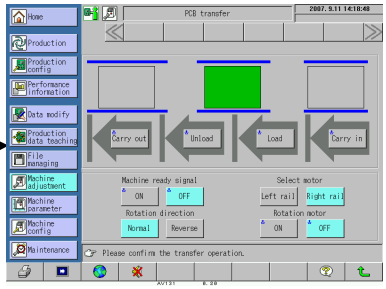




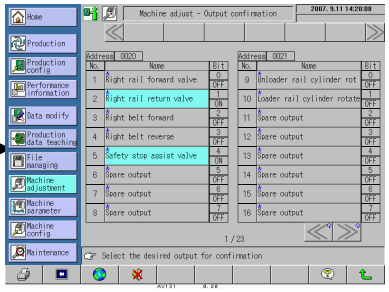
● NC Move



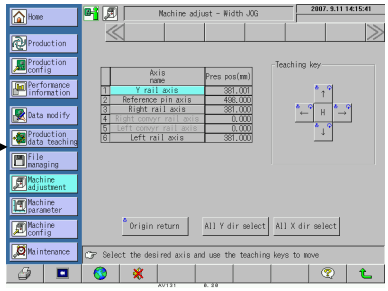
● PCB transfer



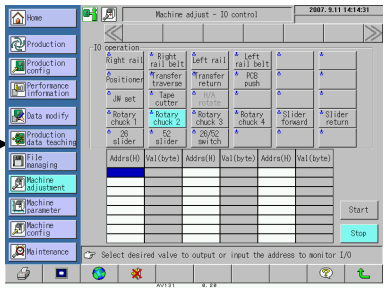
● Output confirmation



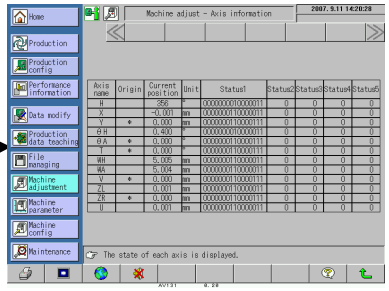
● Width JOG



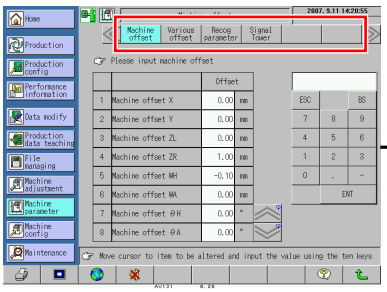
● IO control



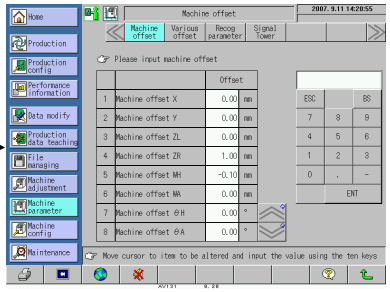
● Axis information



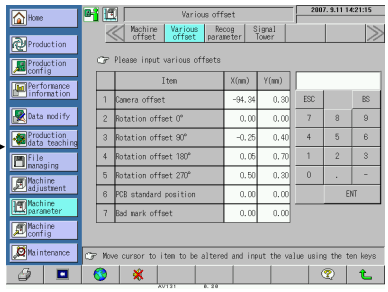
4.1.9 Machine Parameter



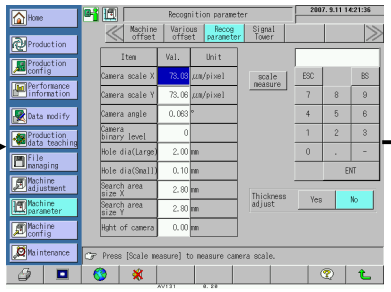
- Machine offset

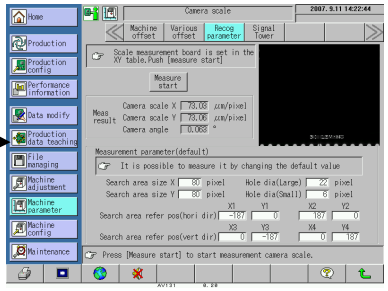


- Various offset

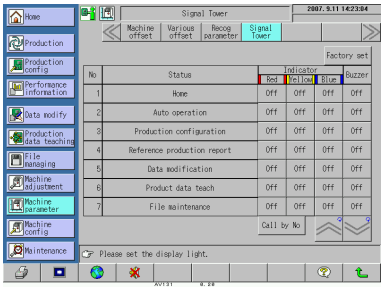


- Recognition parameter



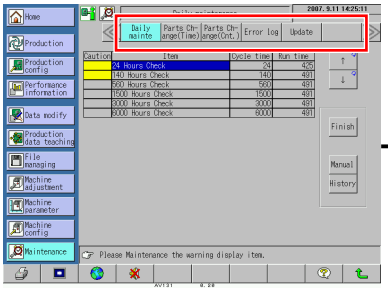


- Signal Tower

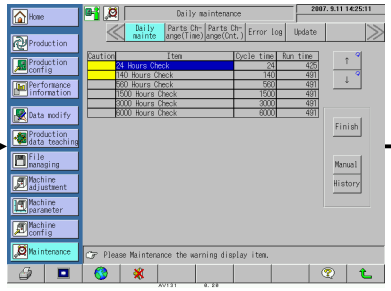


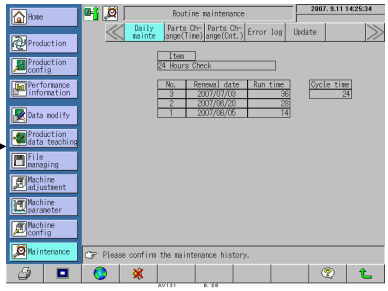
4.1.10 Machine Configuration

4.1.11 Maintenance

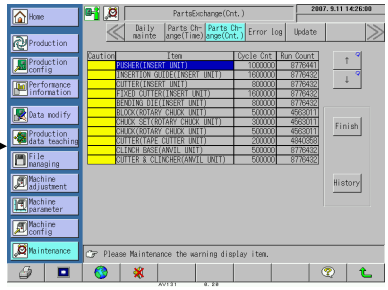


● Daily maintenance

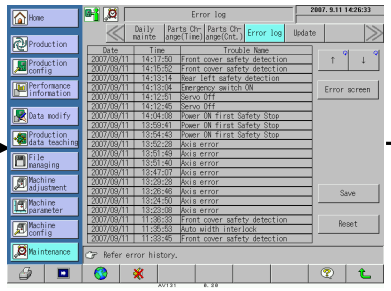




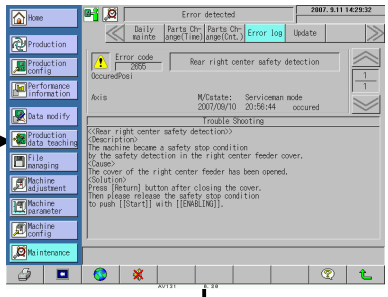
● Parts Change (Count)



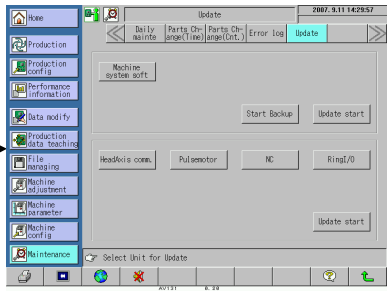
● Error log sub menu



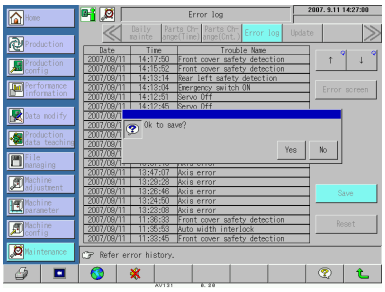
● Error log



● Update



● Save



5. PRODUCTION MANAGEMENT INFORMATION

This chapter describes the method to understand the errors that occur during production and the state of production.

5.1. Performance Information

D79REC-60-010-A0

This function presents the information on the ever-changing conditions of the machine during production. Performance information is divided into the following categories.

- Machine Information
- Model Information (by NC data)
- Data Information (by NC data)
- Feeder Information
- Rotary Chuck Information

■ Cumulative Information:

Information concerning the machine and NC data accumulated since the start of production.
Data are accumulated until cleared.

5.1.1 Data Items on the Performance Information Screen

The data items accumulated on the performance information (Machine information and Data information) screen are summarized as follows.

Data Item	Description	Remarks
Product count	No. of PCBs produced	Unfinished PCBs canceled halfway of NC data are not counted. A produced PCB is counted when the insertion at the EOP step has been completed.
Circuit count	No. of circuits produced	No. of S&R patterns × No. of PCBs
Power on time	Total amount of time machine power has been on	
Operation time	Total amount of time machine has been engaged in production	Counts up while the machine is running in the Production menu mode. Total amount of time from when <START> is pressed until when a 1 step stop, EOP stop, error stop, part exhaust stop or insertion error stop occurs in the Production menu mode.
Operation prepare time	Total amount of time the machine has been stopped with the power on for reasons other than trouble, plus total amount of time in the modes other than Production menu mode.	This is not counted when Maintenance time is being counted.
Maintenance time	Total amount of time from when an error screen is closed until when “START” is pressed.	Counts up from when the error screen is closed in the Production menu mode to when the machine begins operation, as well as to when a mode other than Production menu is selected.
PCB wait (LD) time	Total amount of time spent waiting for PCB on the loader side in the Production menu mode.	Counts up only while a PCB wait error has occurred on the loader side. It is not counted when a PCB wait error has occurred on the unloader side.
PCB wait (UNLD) time	Total amount of time spent waiting for PCB on the unloader side in the Production menu mode.	
Trouble stop	Total amount of time the machine has been stopped due to errors in the Production menu mode.	Counts up while the machine is being stopped with an error message displayed. However, it is not counted in case of part exhaust error and insertion error. Besides, if error occurs in MANUAL or SEMI mode, Operation prepare time is counted.
Insert Error (stop) time	Total amount of time the machine has stopped for insertion errors that have occurred in the Production menu mode.	
Part exhaust (stop) time	Total amount of time the machine has been stopped for part exhaustion errors that have occurred in the Production menu mode.	In case of 120 inputs, if the machine continues running in the exchange mode despite part exhaustion, Part exhaust time is not counted but Operation time is counted.
Total insertion ratio (%)	$\frac{(\text{Insertion count} - \text{Total insertion error count}) \times 100}{\text{Insertion count}}$	
Insertion ratio (%)	$\frac{(\text{Insertion count} - \text{Insertion error count}) \times 100}{\text{Insertion count}}$	

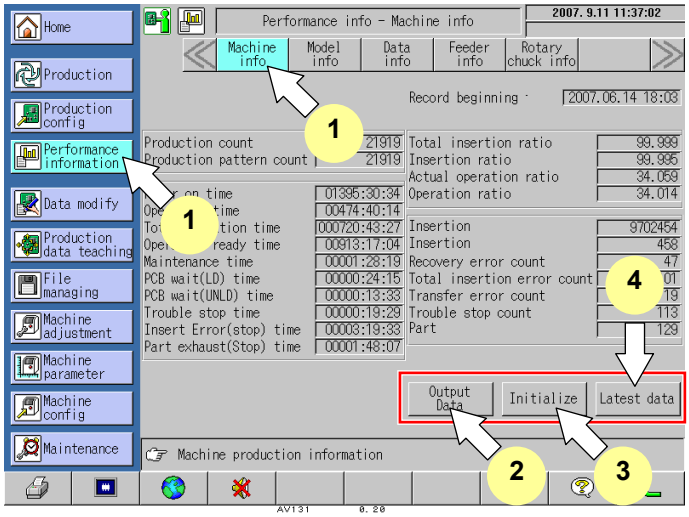
Data Item	Contents	Remarks
Move setting ratio (%)	$\frac{\text{Operation time} \times 100}{\text{Power on time}}$	Rounded to 3 decimal places. However, 99.9995 to 99.9999 will be treated as 99.999%.
OpeRate (%)	$\frac{(\text{Operation time} + \text{PCB wait (LD) time} + \text{PCB wait (UNLD) time}) \times 100}{\text{Power on time}}$	
	Rounded to 3 decimal places. However, 99.9995 to 99.9999 will be treated as 99.999%.	
Insertion count	No of times a part has been inserted successfully.	Counts up when a part inserted has been determined to be OK. Counts up also when the recovery for an insertion error has been determined to be OK.
Insertion error count	No. of insertion errors during normal production	Insertion errors during recovery insertion are not
Recovery error count	No. of insertion errors during recovery insertion	Counts up all the insertion errors until recovery insertion is OK.
Total insertion error count	No. of times recovery insertion has been canceled after an insertion error (insertion cancel).	Counts up when, after the [Recovery] button shown in light blue due to an insertion error has been pressed to turn it off, the recovery insertion has been canceled by the <START> switch.
Transfer error count	No. of times the machine has stopped due to transfer errors	
Trouble stop count	No. of times the machine has stopped due to errors other than insertion errors and part exhaust errors.	Does not count PCB waiting error.
Part exhaust count	No. of times the machine has stopped due to a part exhaust error.	Counts up even when a part has been exhausted but the machine continues with the with the other Z axis table in the exchange mode. When the Z alternate function is used, this is counted when the last spare feeder is used up.
Total operation time	Cumulative operation time. (Same as the hour meter information of the conventional models.)	Turning the power off or initializing the production management information will not clear total operation time. (Machine information)

5.1.2 Performance Information (Machine)

Used to display, output and initialize the machine information accumulated from the start of production.

◆ How to display/output/clear the machine information

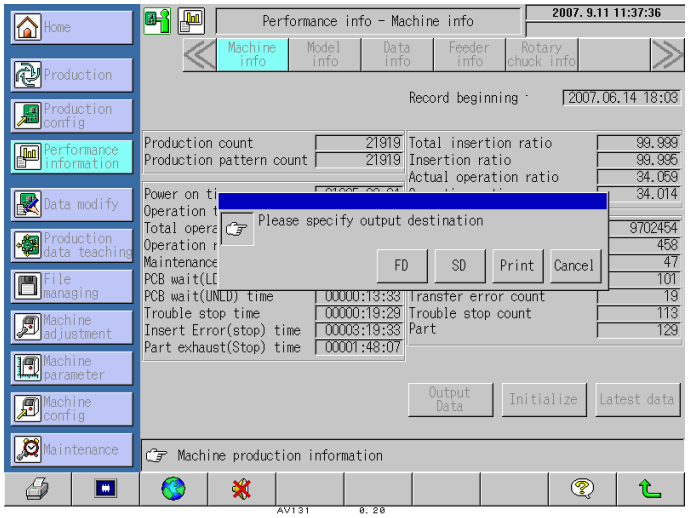
1. Press [Performance information] on the Main menu followed by [Machine info] on the sub-menu to display the machine information.



2. Outputting data

- Press [Output Data] to save the machine information file to FD or SD.
  Press [Print] to print out the information to the AV131's built-in printer. (Optional)
  =HINT=
  Use the function to save the data periodically before clearing the performance information.

- Press [Cancel] to cancel outputting information.



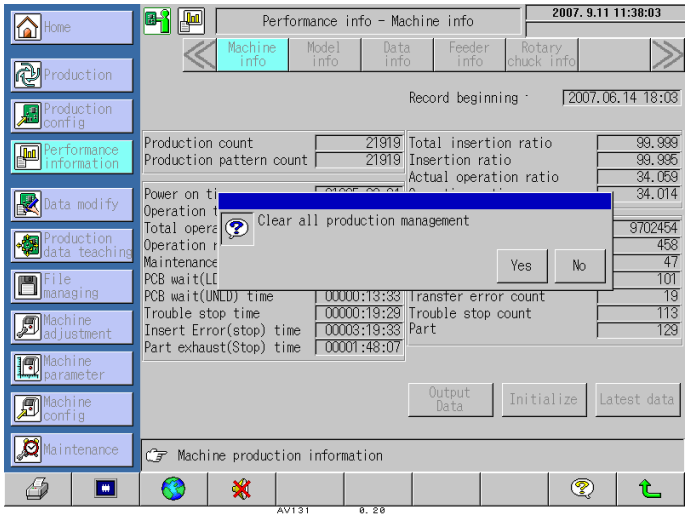
3. Initializing Data

- Used to clear all cumulative performance information of machine (displayed on the screen).

(1) Pressing [Initialize] displays the confirmation dialog box.
(2) Press [Yes] to clear the data.
=REMARKS=
Data cannot be initialized while the machine is engaged in production.
(3) Press [Cancel] to cancel initializing information.

4. Updating data

- Press [Latest data] to update the machine information data.

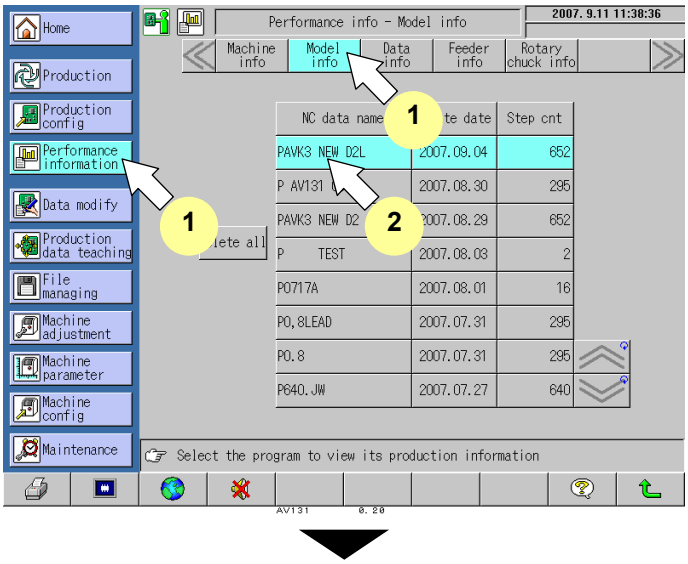


5.1.3 Performance Information (Model)

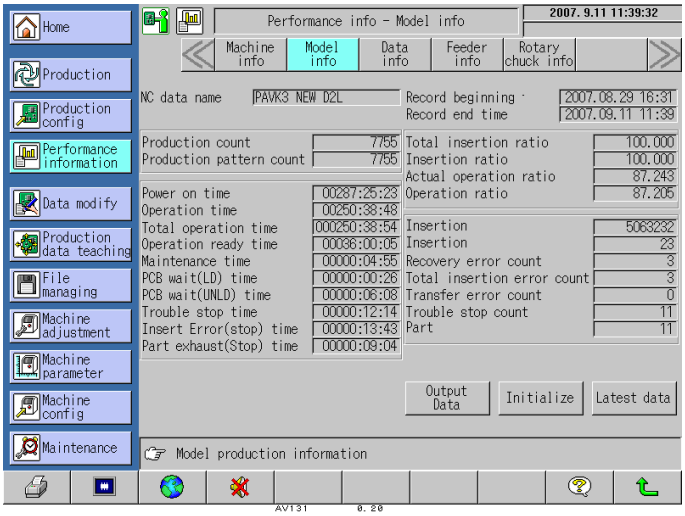
Used to display, output and initialize the selected NC data information accumulated from the start of production.

◆ How to display/output/clear the model information

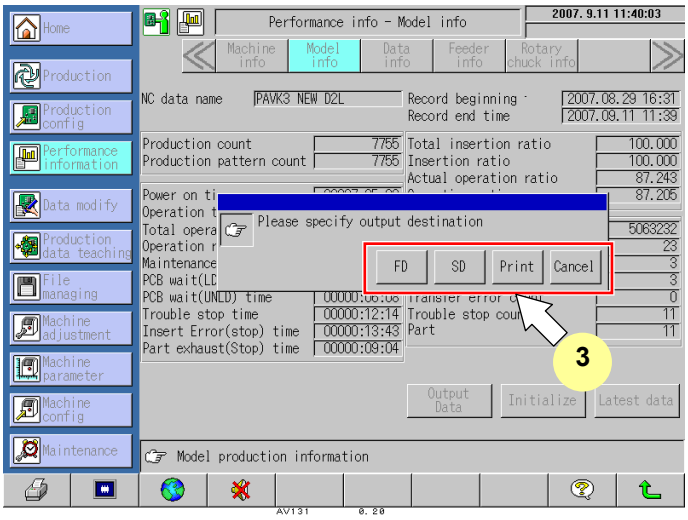
1. Press [Performance information] on the Main menu followed by [Model info] on the sub-menu to display the list of the NC data.



2. Select the target NC data name to display its model information.



3. Outputting data
   - Press [Output Data] to save the model information file to FD or SD.
     Press [Print] to print out the data to the AV131's built-in printer. (Optional)
     =HINT=
     Use the function to save the data periodically before clearing the performance information.
   - Press [Cancel] to cancel outputting information.



4. Initializing data

- Used to clear all cumulative performance information of the selected NC data (displayed on the screen).

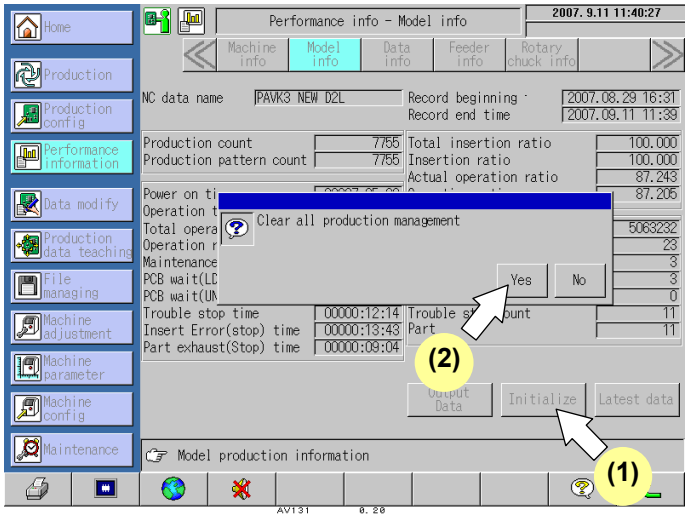
(1) Pressing [Initialize] displays the confirmation dialog box.

(2) Press [Yes] to clear the data.

=REMARKS=

Data cannot be initialized while the machine is engaged in production.

(3) Press [No] to cancel initializing information.



5. Deleting all data

- Used to clear all cumulative performance information of the selected NC data.

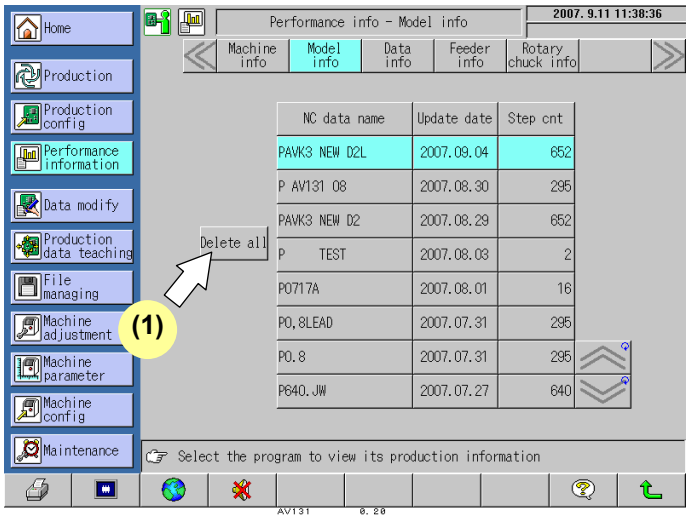
(1) Press [Delete all] at the left of the NC data list.

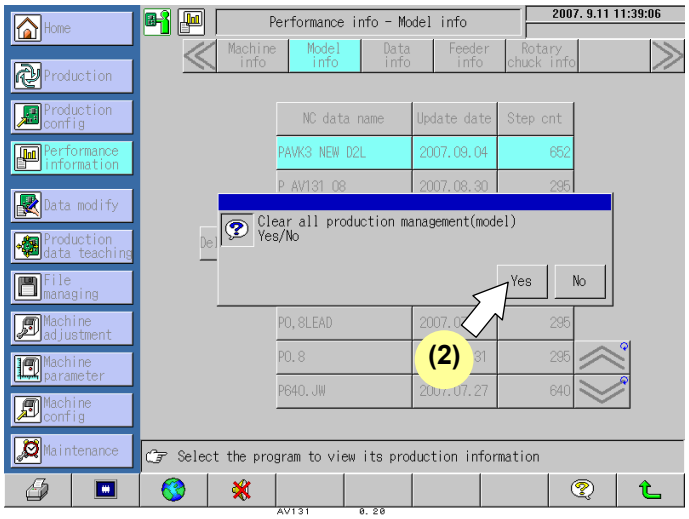
(2) Press [Yes] to clear the all data.

=REMARKS=

Data cannot be deleted while the machine is engaged in production.

(3) Press [No] to cancel deleting all information.





5.2. NC Data Information

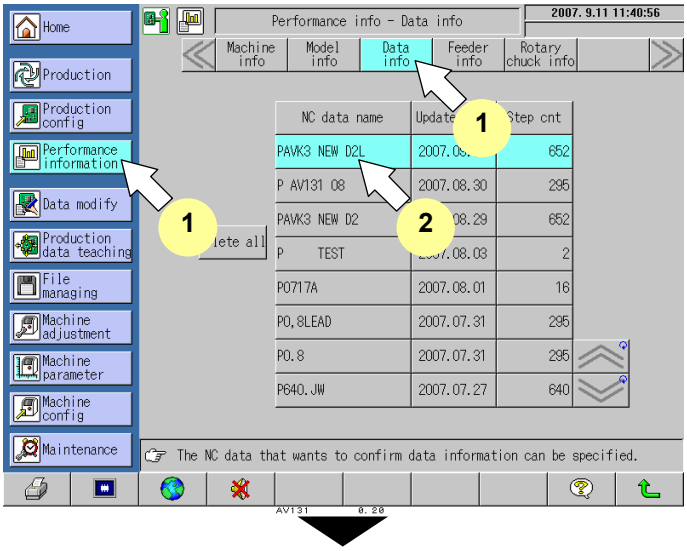
D79REC-60-150-A0

5.2.1 Data Information

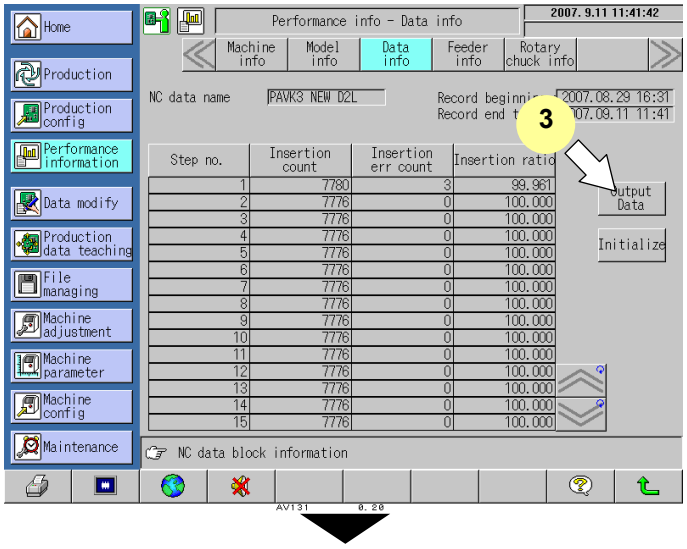
The NC Data Information function is used to display, output, and initialize insertion data at each individual step of NC data accumulated from the start of production.

◆ Data Information

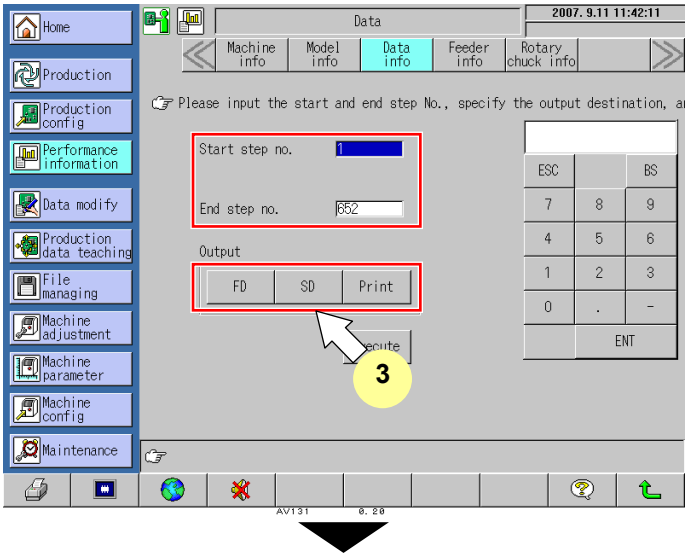
1. Press [Performance information] on the Main menu followed by [Data info] on the sub-menu to display the list of the NC data.



2. Select the target NC data name to display its data information.



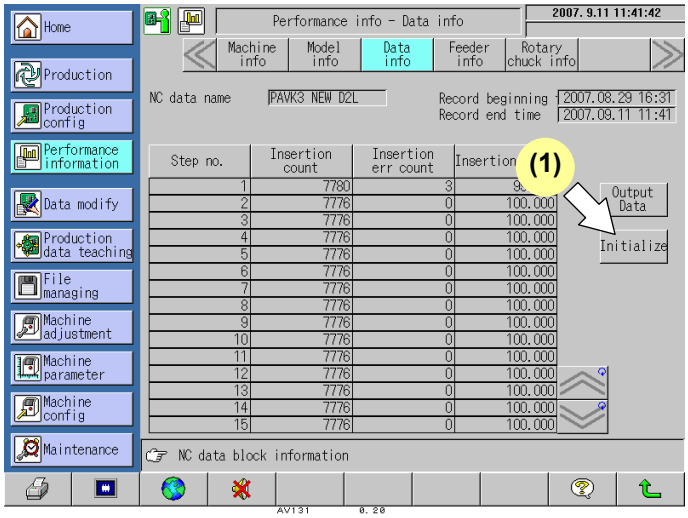
3. Outputting data
   - Press [Output Data] and specify where the data file is saved. When FD or SD card is selected, the data file will be saved on the specified medium.
     Press [Print] to print out the data to the AV131's built-in printer. (Optional)
     =HINT=
     Specify the start feeder No. and the finish feeder No. to save the data.

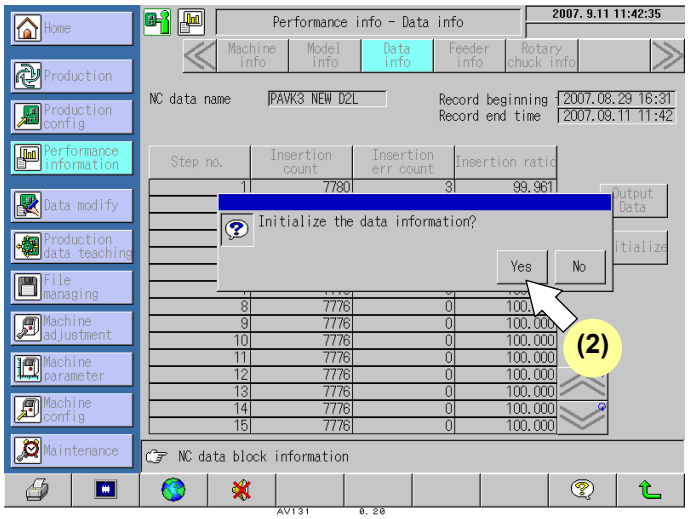


4. Initializing data

- Used to clear all cumulative information of the selected NC data (displayed on the screen).

(1) Pressing [Initialize] displays the confirmation dialog box.
(2) Press [Yes] to initialize the data.

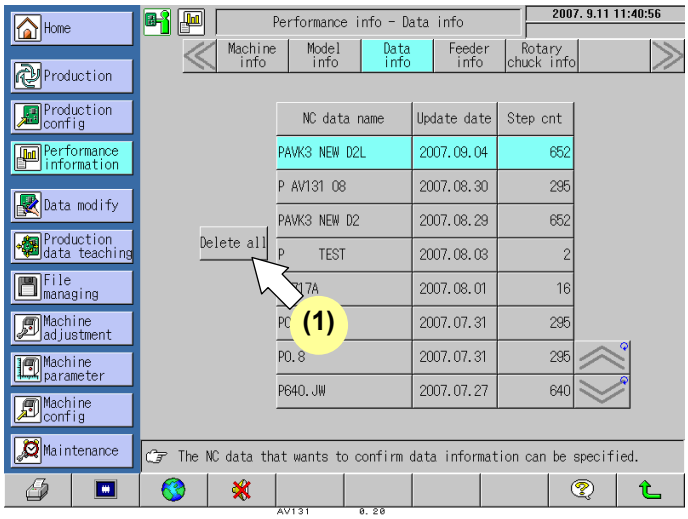


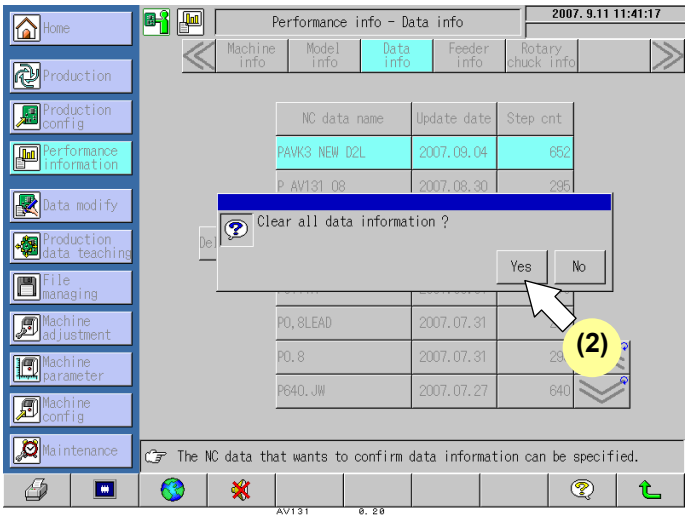


5. Deleting all data

- Used to clear all cumulative data information and performance information of NC program data.

(1) Press [Delete all] at the left of the NC data list.
(2) Press [Yes] to delete all the data.




Home
Production
Production config
Performance information
Data modify
Production data teaching
File managing
Machine adjustment
Machine parameter
Machine config
Maintenance
Performance info - Data info
2007. 9.11 11:41:17
Machine info
Model info
Data info
Feeder info
Rotary chuck info
NC data name
Update date
Step cnt
PAVK3 NEW D2L
2007.09.04
652
P AV131 08
2007.08.30
295
Clear all data information ?
Yes
No
(2)
P0.8LEAD
2007.07.31
P0.8
2007.07.31
P640.JW
2007.07.27
640
The NC data that wants to confirm data information can be specified.
AV131
0.20

5.3. Feeder Information

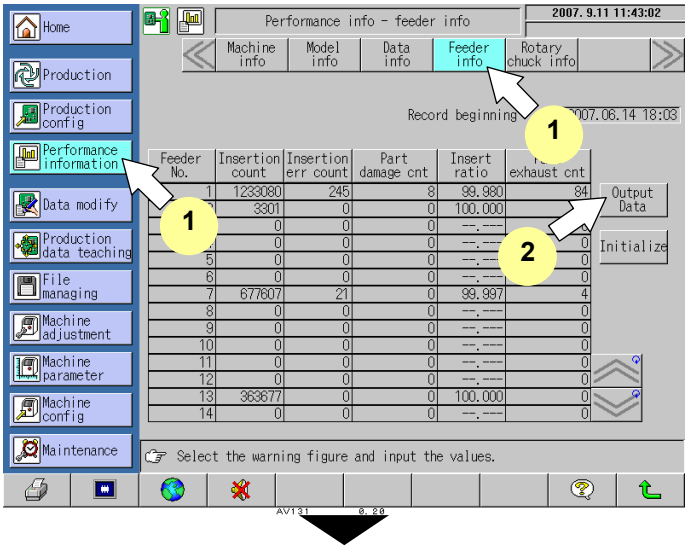
D79REC-60-280-A0

5.3.1 Feeder Information

The Feeder Information function is used to display, output, and initialize information data of each individual feeder of the part supply unit (Z axis).

◆ Feeder Information

1. Press [Performance information] on the Main menu followed by [Feeder info] on the sub-menu to display the list of the feeder information.

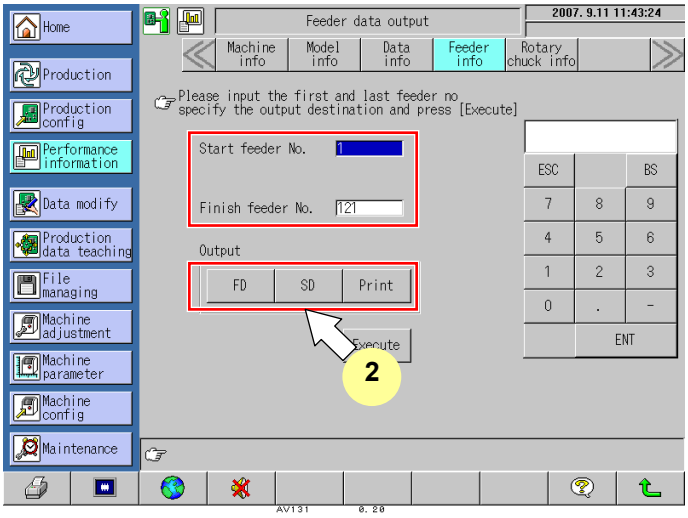


2. Outputting data

- Press [Output Data] and specify where the data file is saved. When FD or SD card is selected, the data file will be saved on the specified medium.
  Press [Print] to print out the data to the AV131's built-in printer. (Optional)

  =HINT=

  Specify the start feeder No. and the finish feeder No. to save the data.

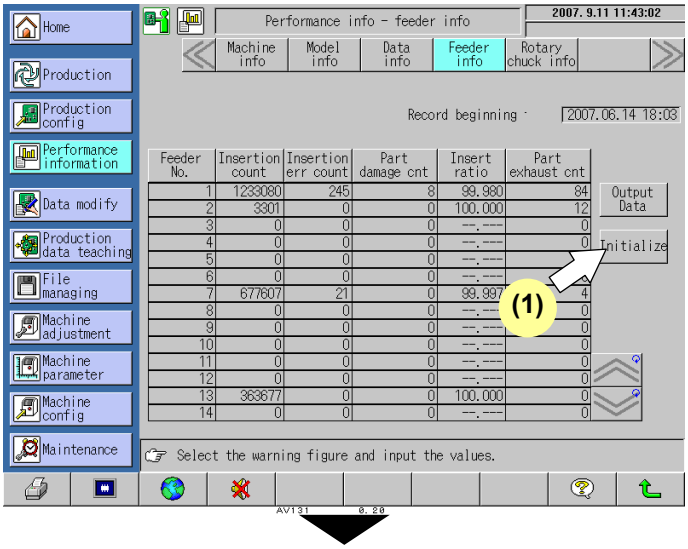


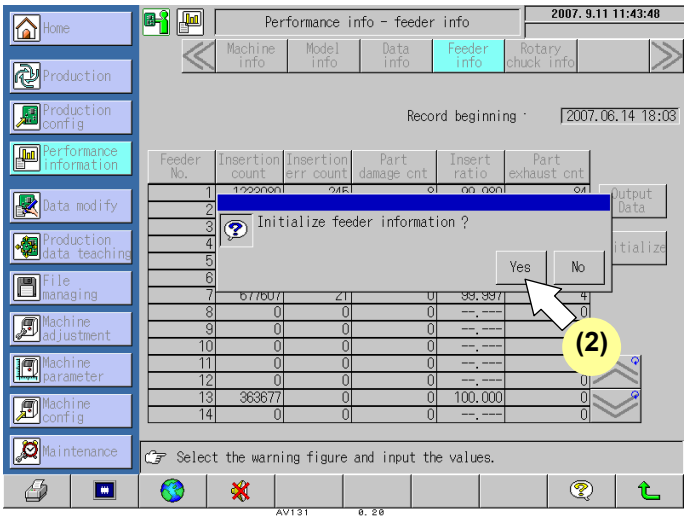
3. Initializing Data

- Used to clear all feeder information (displayed on the screen).

(1) Pressing [Initialize] displays the confirmation dialog box.

(2) Press [Yes] to initialize the data.




Home
Production
Production config
Performance information
Data modify
Production data teaching
File managing
Machine adjustment
Machine parameter
Machine config
Maintenance
Performance info - feeder info
2007. 9.11 11:43:48
Machine info
Model info
Data info
Feeder info
Rotary chuck info
Record beginning :
2007.06.14 18:03
Feeder No.
Insertion count
Insertion err count
Part damage cnt
Insert ratio
Part exhaust cnt
Output Data
Initialize
Initialize feeder information ?
Yes
No
(2)
Select the warning figure and input the values.

5.4. Rotary Chuck Information

D79REC-60-190-A0

5.4.1 Data Items on the Rotary Chuck Information Screen

The data items accumulated in the rotary chuck information are as follows.

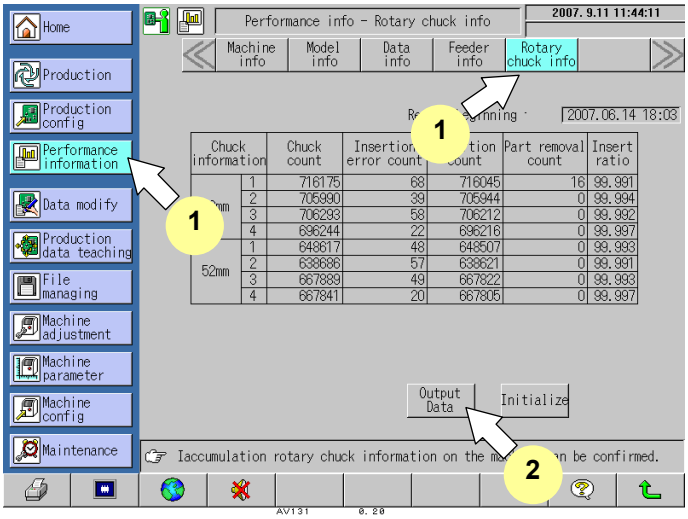
Data Item	Description	Remarks
Chuck count	No. of times parts chucked from the feeder (Chuck frequency)	Regardless of whether the part detection sensor is provided.
Insertion error count	No. of times an insertion error occurred.	Including recovery insertion errors.
Insert count	No. of times a part was inserted successfully.	
Insert ratio (%)	$\frac{(\text{Insertion count-Insertion error count}) \times 100}{\text{Insertion count}}$	

5.4.2 Displaying the Rotary Chuck Information

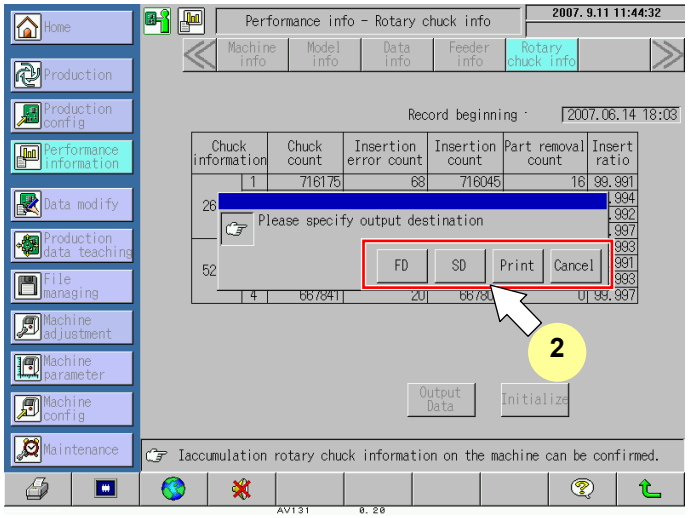
Used to display, output and initialize the production information data of individual rotary chuck.

◆ Rotary Chuck Information

1. Press [Performance information] on the Main menu followed by [Rotary chuck info] on the sub-menu to display the rotary chuck information.

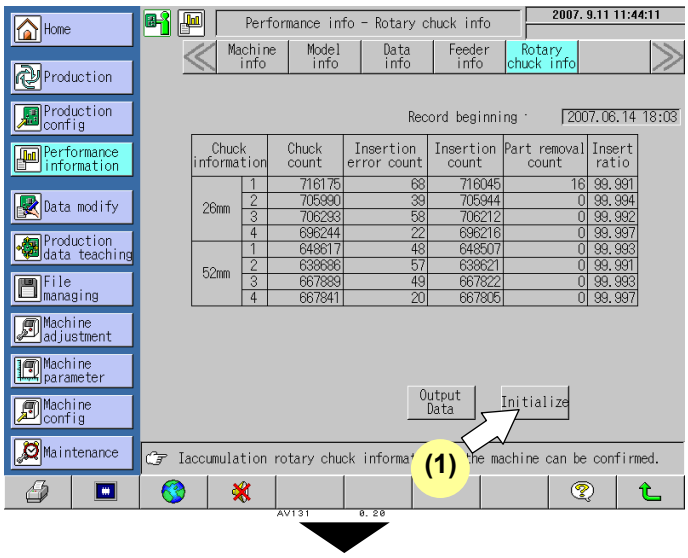


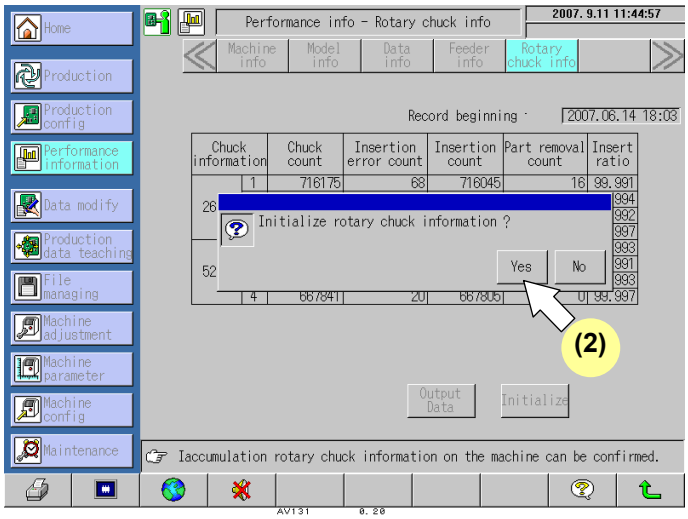
2. Outputting data
   - Press [Output Data] and specify where the data file is saved. When FD or SD card is selected, the data file will be saved on the specified medium.
     Press [Print] to print out the data to the AV131's built-in printer. (Optional)
     =HINT=
     Specify the start feeder No. and the finish feeder No. to save the data.



3. Initializing data
   - Used to clear all cumulative rotary chuck information (displayed on the screen).

   (1) Press [Initialize] at the left of the NC data list.
   (2) Press [Yes] to initialize the data.




Home
Production
Production config
Performance information
Data modify
Production data teaching
File managing
Machine adjustment
Machine parameter
Machine config
Maintenance
Performance info - Rotary chuck info
2007. 9.11 11:44:57
Machine info
Model info
Data info
Feeder info
Rotary chuck info
Record beginning
2007.06.14 18:03
Chuck information
Chuck count
Insertion error count
Insertion count
Part removal count
Insert ratio
Initialize rotary chuck information ?
Yes
No
(2)
Output Data
Initialize
Iaccumulation rotary chuck information on the machine can be confirmed.

6. SYSTEM MANAGEMENT

This chapter describes the data setting for using the machine and the manual operation methods.

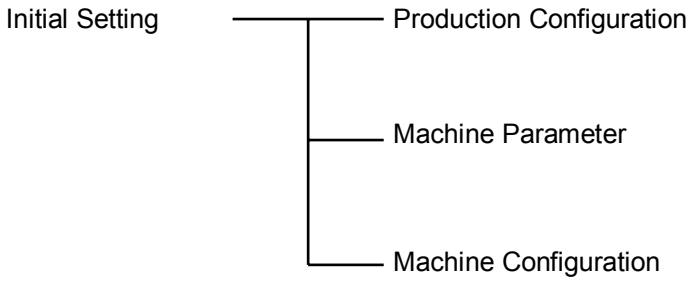
6.1. Initial Setting (Overview)

D79REC-70-010-A0

6.1.1 Overview

Positional relations among the units of the AV131 are factory-adjusted as individual data (initial values) before shipment from the factory.
Some of these data can be user-set while some do not require particular setting besides the factory adjustment.

Initial setting includes: ‘Machine Parameter’, ‘Machine Configuration’, and ‘Production Configuration’.

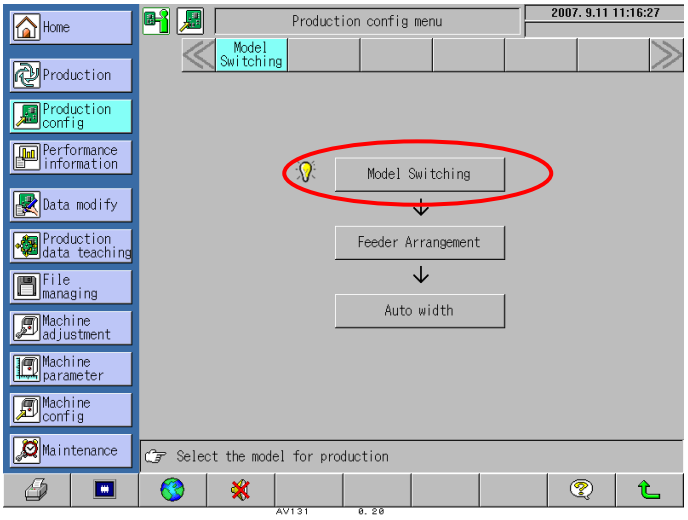


6.2. Production Configuration

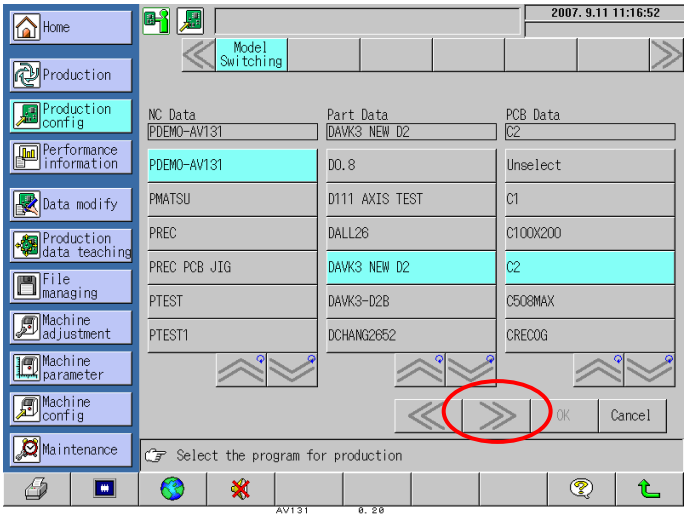
D79REC-70-670-A0

The Production Configuration menu includes the production condition data to be set when selecting the production data.

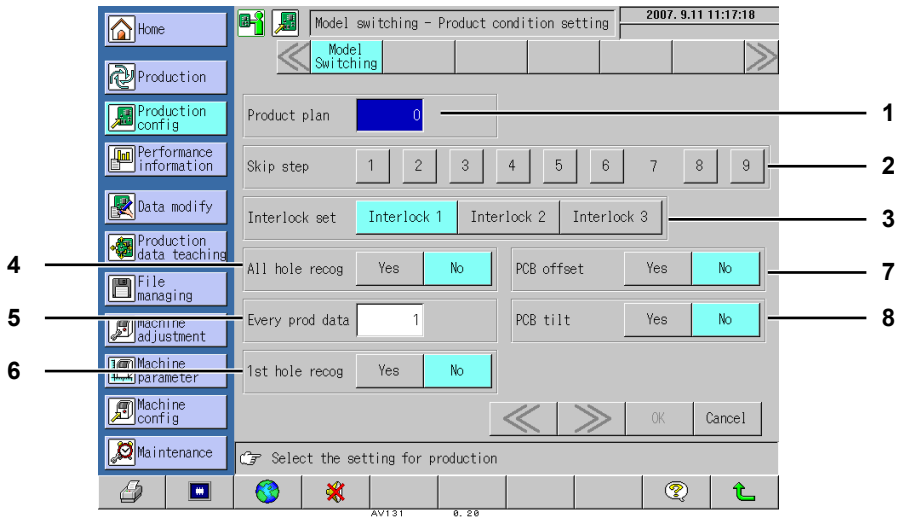
1. Press [Production Configuration] on Main Menu to call up the Model Switching Manu screen.



2. Pressing the [Model switching] changes the display to the screen for selecting the production data. Select a desired data name from the respective lists (NC data, part data and PCB data), and press among the buttons below the lists.



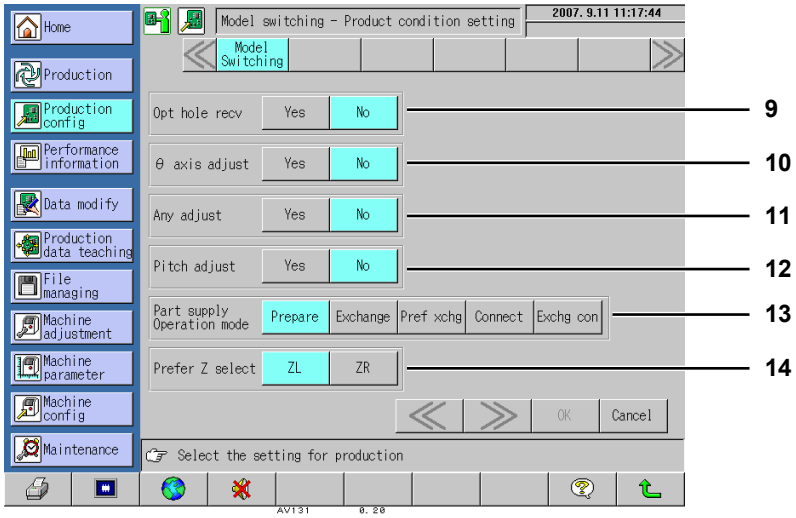
3. Product Condition Setting screen is displayed. Conditions can be set by selecting the items and inputting data via this screen.



1	Prod plan count	Set the number of PC boards to be produced with the selected NC data. **=REFERENCE=** Setting the count as zero is equivalent to not setting the count.
2	Skip step	Set the step to be skipped in the selected NC data. More than one skip step can be designated. **=REFERENCE=** Skip step 7 is default setting (Unconditioned skip).
3	Interlock setting	Set the interlock mode.
	(1) Interlock 1	Set this item when there are tall component already placed on the target PCB. **=REFERENCE=** (Factory setting is Interlock 1.)
	(2) Interlock 2	Set this item when there is no tall component already placed on the target PCB. **=REFERENCE=** (This setting makes X-Y table move for a longer distance within the tact time than Interlock 1.)
	(3) Interlock 3	This is a setting when an axial part to be inserted is tall although there is not tall part on PCB. Interlock 2 is used for JW insertion and interlock 1 for axial component insertion.

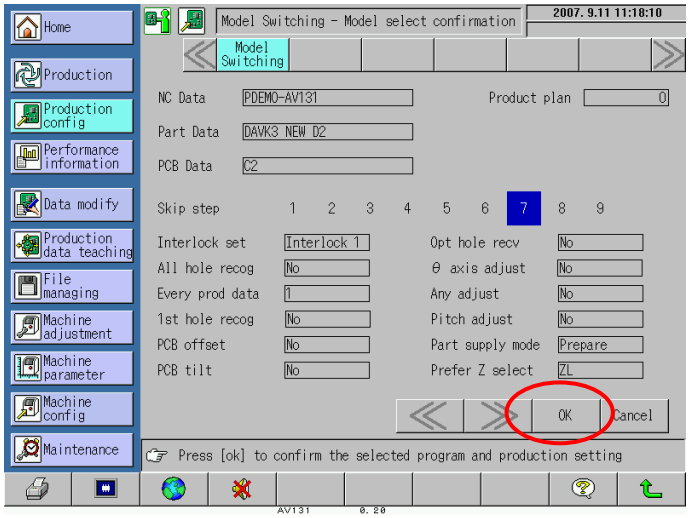
=REMARKS=
For the criteria for tall components, refer to the specification description.

4	All holes recognition	This function is used to recognize all insertion holes before producing PCBs, and so as to take the displacement of insertion points into account. **= REFERENCE =** When the setting is enabled, the displacements in the result are held until the next time all holes recognition is performed.
5	Every production data	When All hole recog is enabled, set the number of desired PCBs. All holes recognition will be executed each time the set count is reached.
6	1st hole recognition	This function recognizes the 1st insertion hole of the NC program, and based on that result, it executes offset for all insertion holes on PCB. In case of S&R data, the first insertion hole of each pattern is recognized and the rest of holes on each pattern are corrected based on that result.
7	PCB offset	This function recognizes one of the reference hole that were taught, and corrects the parallel displacement of the entire PCB.
8	PCB tilt	This function recognizes both of the reference holes that were taught, and corrects the angular displacement of the entire PCB and the shifts of all insertion holes due to the angular displacement.



9	Opt hole recovery	This function recognizes the hole position when performing recovery after an insertion error to correct any displacement.
10	θaxis adjust	This function recognizes the positions of the two holes at the insertion point to adjust any rotational displacement of the hole.
11	Any adjust	This function recognizes only the specified insertion holes to correct any displacement of the holes. **= REFERENCE =** Insertion point to be recognized needs to be designated at the NC Data Editing.
12	Pitch adjustment	This function recognizes the positions of the two holes at the insertion point to adjust the insertion pitch.
13	Part supply Operation mode	This is to designate the operation mode of the part supply.
14	Prefer Z select	Set the Z axis table to be used preferentially in the Pref connect mode.

4. When all the settings are completed, press the [≫] and the Model select confirmation screen is displayed.



5. This screen is to check all the items that are set at the production configuration.

- If all the set items are correct, press the [OK] to determine the settings.
- Pressing the [Cancel] discards any changes made. (The previous settings remain unchanged.)
- To change the settings again, press the [≪] to return to the setting screen.

6.3. Machine Parameter

D79REC-70-570-A0

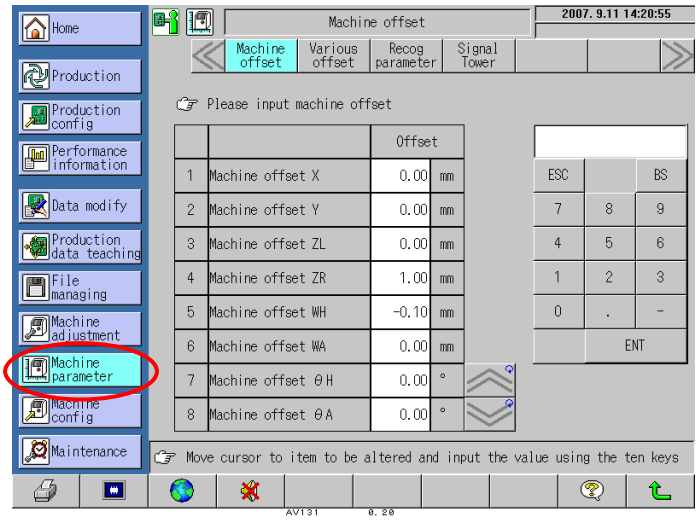
Machine parameter menu sets the position in relation of each part in AV131.

=REMARKS=

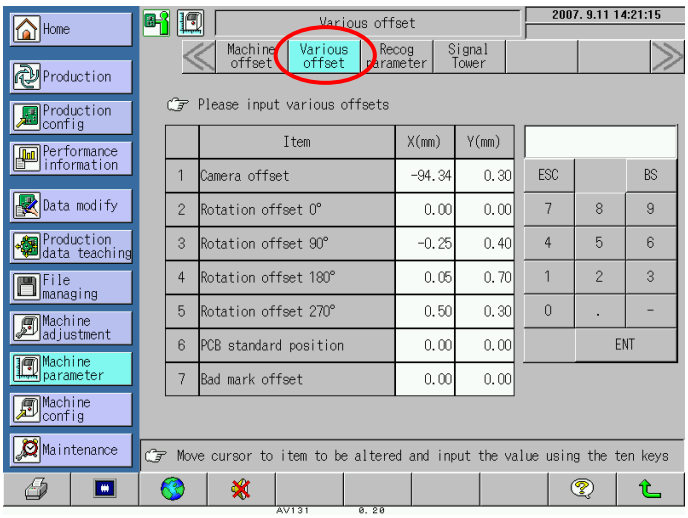
The parameters are factory adjusted. Before making any changes, familiarize yourself with the setting items, and use extreme care when actually changing the setting.

It is also recommended to save the machine data before making changes so that they can be restored to the original settings.

1. Press [Machine Parameter] on the Main Menu which brings the Machine Offset screen.

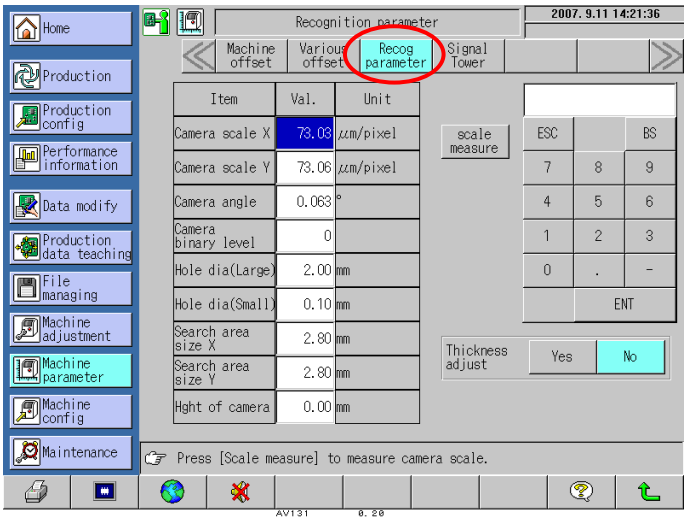


2. Press [Various offset] on the sub menu to show the setting screen for camera offset, etc.



Rotation offset 0°: X-Y data offset at 0° insertion
Rotation offset 90°: X-Y data offset at 90° insertion
Rotation offset 180°: X-Y data offset at 180° insertion
Rotation offset 270°: X-Y data offset at 270° insertion

3. Similarly, pressing [Recog parameter] on the sub menu calls up the setting screen for camera scale, etc.

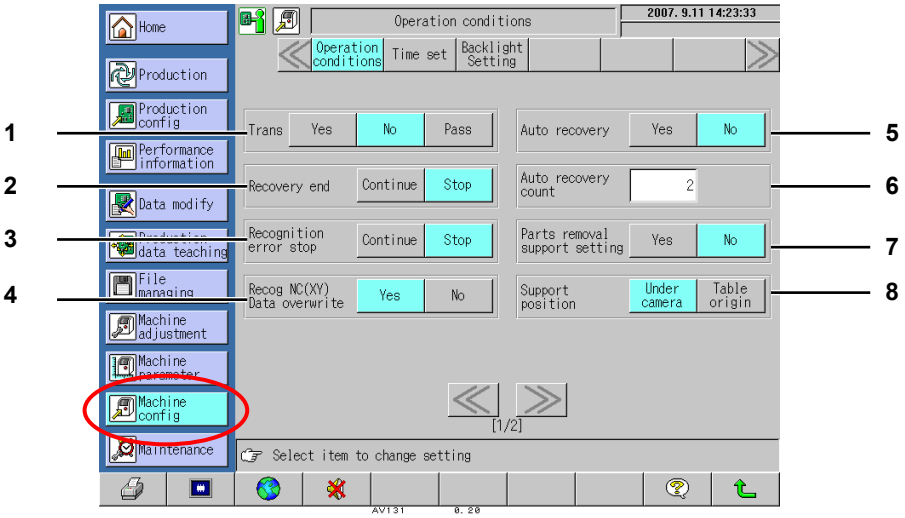


6.4. Machine Configuration

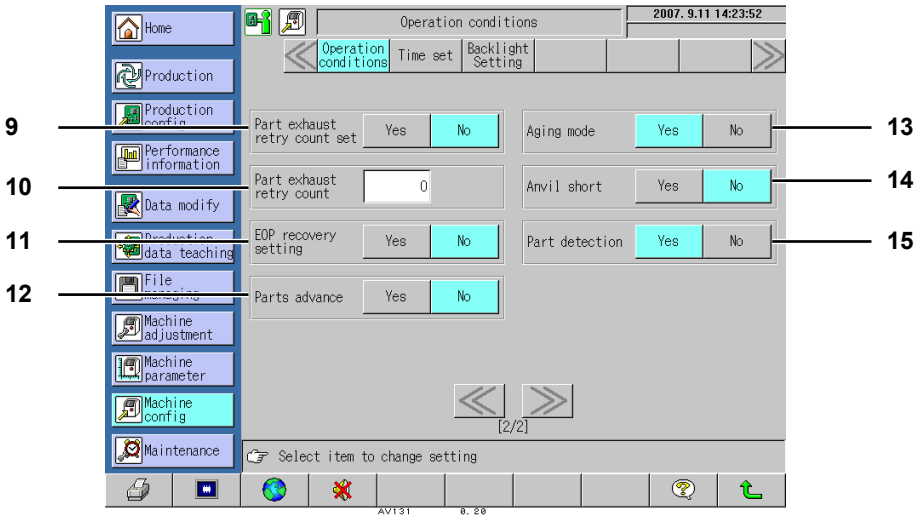
D79REC-70-680-A0

Machine configuration is to set the operation conditions of AV131.

1. Pressing [Machine config] on Main Menu calls up the Operation conditions setting screen.



1	Trans	Sets the PCB transfer.
	(1) Yes	Transfers PCBs.
	(2) No	Does not transfer PCBs.
	(3) Pass	Uses the machine as a conveyor.
2	Recovery end	Sets whether to continue operation without stopping the machine or to stop the machine for checking insertion if the recovery for an insertion error occurs.
3	Recognition error stop	Sets whether to stop the machine or to continue operation if a recognition error occurs.
4	Recog NC (XY) data over write	Sets whether or not to overwrite the XY coordinates in the NC data based on the recognition result when all holes recognition is enabled.
5	Auto recovery	Sets whether or not to use auto recovery function for an insertion error. 'Yes': Used 'No': Not used
6	Auto recovery count	Sets the maximum number of times to reattempt auto recovery in case of recovery for an insertion error.
7	Parts removal support setting	Sets whether or not to enable the function for moving the X-Y table to the position where it is easy to remove the component after an insertion error.
8	Support position	Sets the position to which the X-Y table moves to when Parts removal support setting is enabled.
	(1) Under camera	When an insertion error occurs, the X-Y table moves to under the camera.
	(2) Table origin	When an insertion error occurs, the X-Y table moves to the origin.



9	Part exhaust retry count set	Sets whether or not to use the Part exhaust retry count set function. 'Yes': Used 'No': Not used
10	Part exhaust retry count	Sets the maximum number of times to reattempt part exhaust retries.
11	EOP recovery setting	Sets whether or not to use the EOP recovery setting function. 'Yes': Used 'No': Not used
12	Parts advance	Sets whether or not to use the Parts advance function. 'Yes': Used 'No': Not used
13	Aging mode	Sets the aging mode. **=Reference=** Aging mode refers to an operation mode which executes production operation without inserting compoments.
14	Anvil short	Sets whether or not to enable the anvil short function.
15	Part detection	Sets whether or not to enable the part detection function **=Reference=** If the part detection is disabled, part detection always results in OK.

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